Properties and Partition Equilibria of Organic Compounds in the Environment

Prof. Dr. Tobias Licha

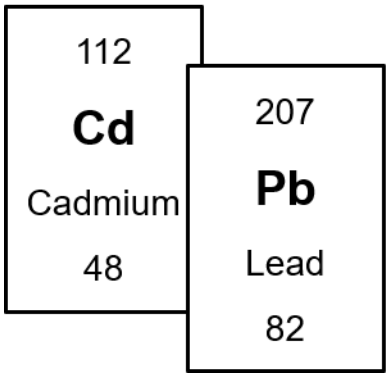
Dr. Jens Prothmann

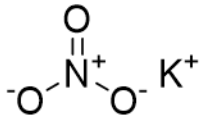
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BOCHUM

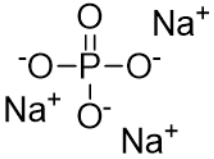
RUB

Inorganic and organic compounds

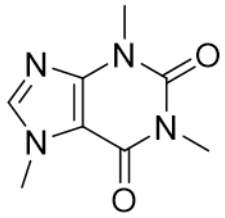
Inorganic compounds

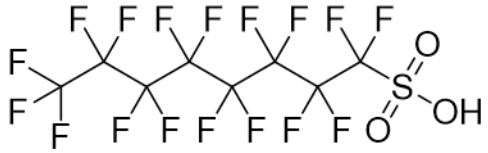


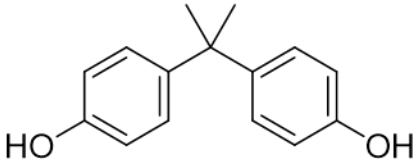


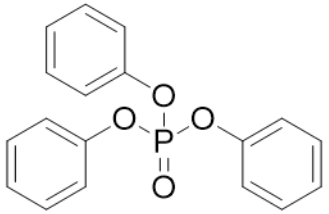


Organic compounds



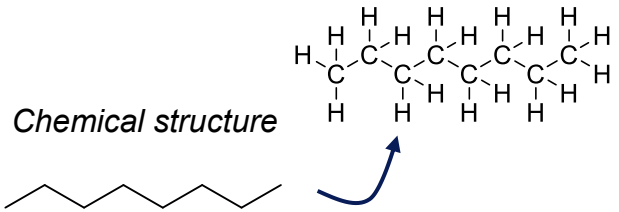
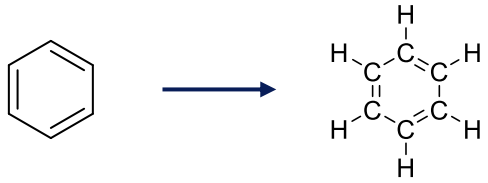
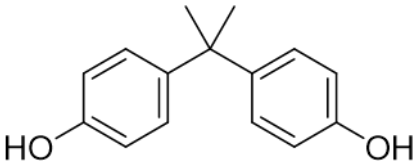






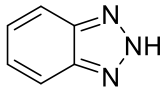
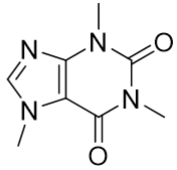
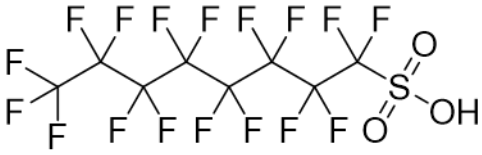
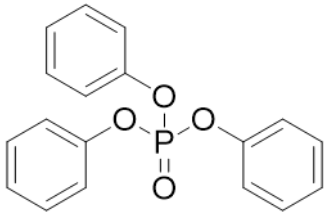
Organic compounds

Groups of organic compounds depending on elemental composition and chemical structure

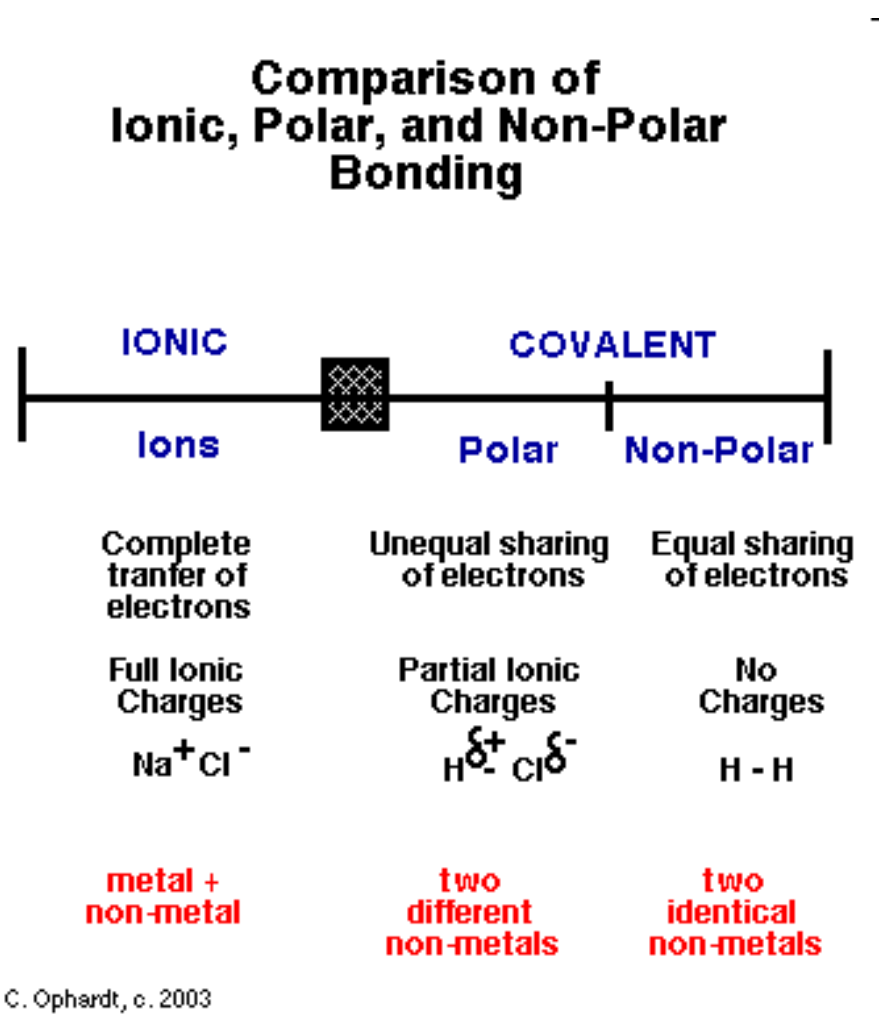
Group name	*Chemical Formula*		*Chemical structure*
Aliphatic hydrocarbons	C_8H_{18}		
Aromatic hydrocarbons	C_6H_6		
Oxygen-containing organic compounds	$C_5H_{10}O_2$	$C_5H_{12}O$	
Aromatic oxygen-containing organic compounds	$C_{15}H_{16}O_2$		

Organic compounds

Groups of organic compounds depending on elemental composition and chemical structure

Nitrogen-containing organic compounds	$C_6H_5N_3$	
Nitrogen+oxygen-containing organic compounds	$C_6H_5N_3$	
Halogenated organic compounds	$C_8H_1O_3S_1F_{17}$	
Phosphor-containing organic compounds	$C_{18}H_{15}O_4P$	

Polarity



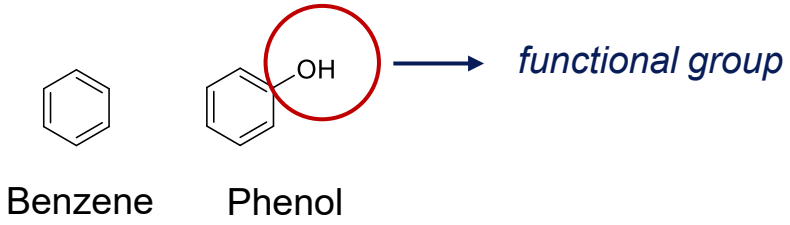
Occurs in molecules were covalent bonds are realised by partners of different electro negativity

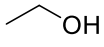
Affects:

- **Melting/Boiling point**
- **Volatility**
- **Solubility**

Organic compounds

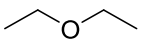
Compound	Water solubility at 25 °C [g/L]	Density [g/cm³]	Boiling point [°C]
Benzene	1.8	0.88	80.1
Phenol	82	1.07	182



Alcohol 

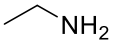
Ketone 

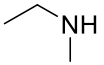
Carboxylic acid 

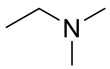
Ether 

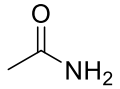
Alkyl chloride 

Alkyl fluoride 

Amine 

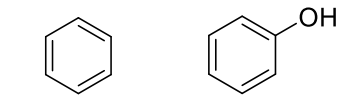
Secondary amine 

Tertiary amine 

Amide 

Volatile and non-volatile organic compounds

Compound	Water solubility at 25 °C [g/L]	Density [g/cm³]	Boiling point [°C]	Henry's law volatility constant [atm·m³/mol]
Benzene	1.8	0.88	80.1	5.9×10^{-3}
Phenol	82	1.07	182	3.0×10^{-7}



Benzene Phenol

Compound	Acronym	Volatilization rate	Henry's law volatility constant [atm m³/mol]
Highly volatile organic compounds	HVOCs	Rapid	$>10^{-3}$
Volatile organic compounds	VOCs	Significant	10^{-5} to 10^{-3}
Semi-volatile organic compounds	SVOCs	Slow	10^{-7} to 10^{-5}
Non-volatile organic compound	NVOCs	Not important	$<10^{-7}$

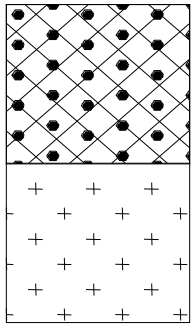
Octanol-Water-Partition coefficient (measure of polarity)

Most important parameter in organic hydrochemistry

BUT: Only valid for uncharged (=non-ionic) organic compounds

Based on *absorption*

$$K_{OW} = \frac{C_O}{C_W}$$



Octanol with dissolved organic compound

Water with dissolved organic compound

K_{OW} = Octanol-Water Partition Coefficient (-)

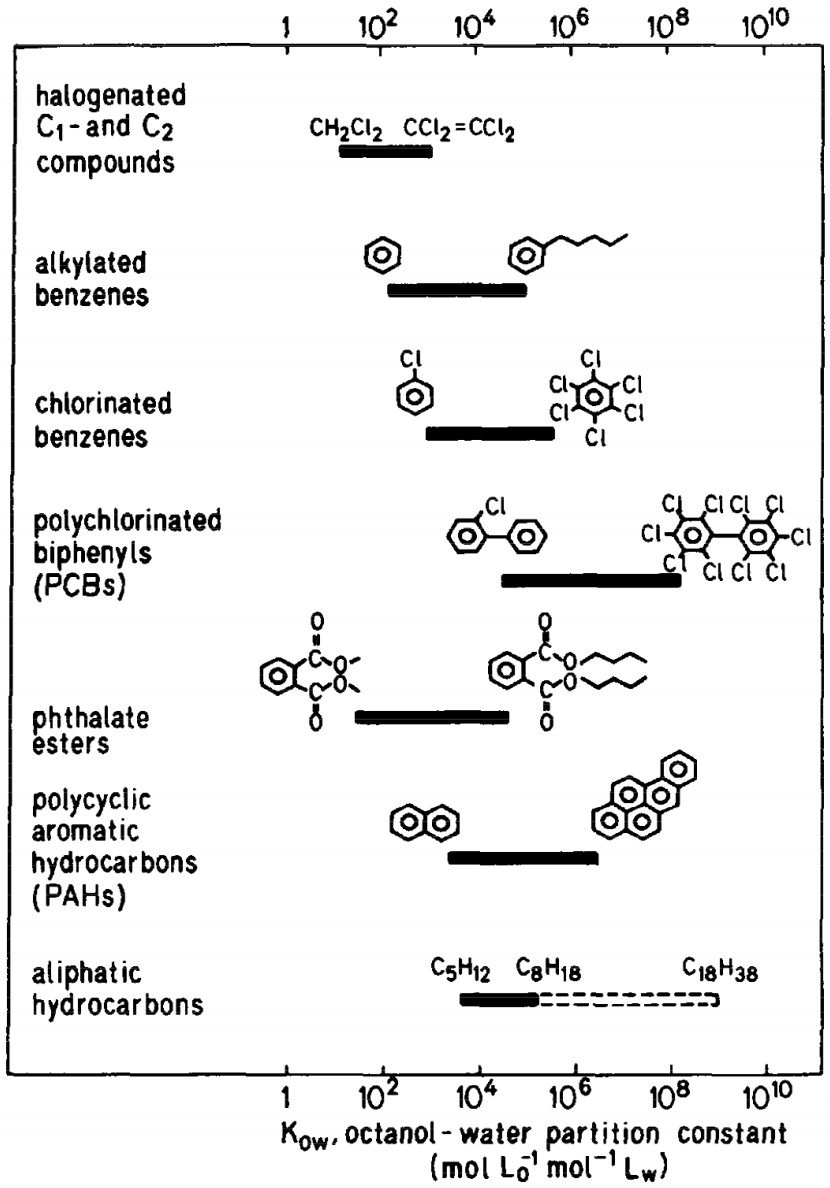
C_O = Concentration of the compound in octanol (mol m^{-3})

C_W = Concentration of the compound in water (mol m^{-3})

Octanol is partially mixable with water (PMOS).

Water-saturated octanol contains 21 mol% water, octanol-saturated water contains 0.008 mol% octanol.

K_{OW} – compound classes

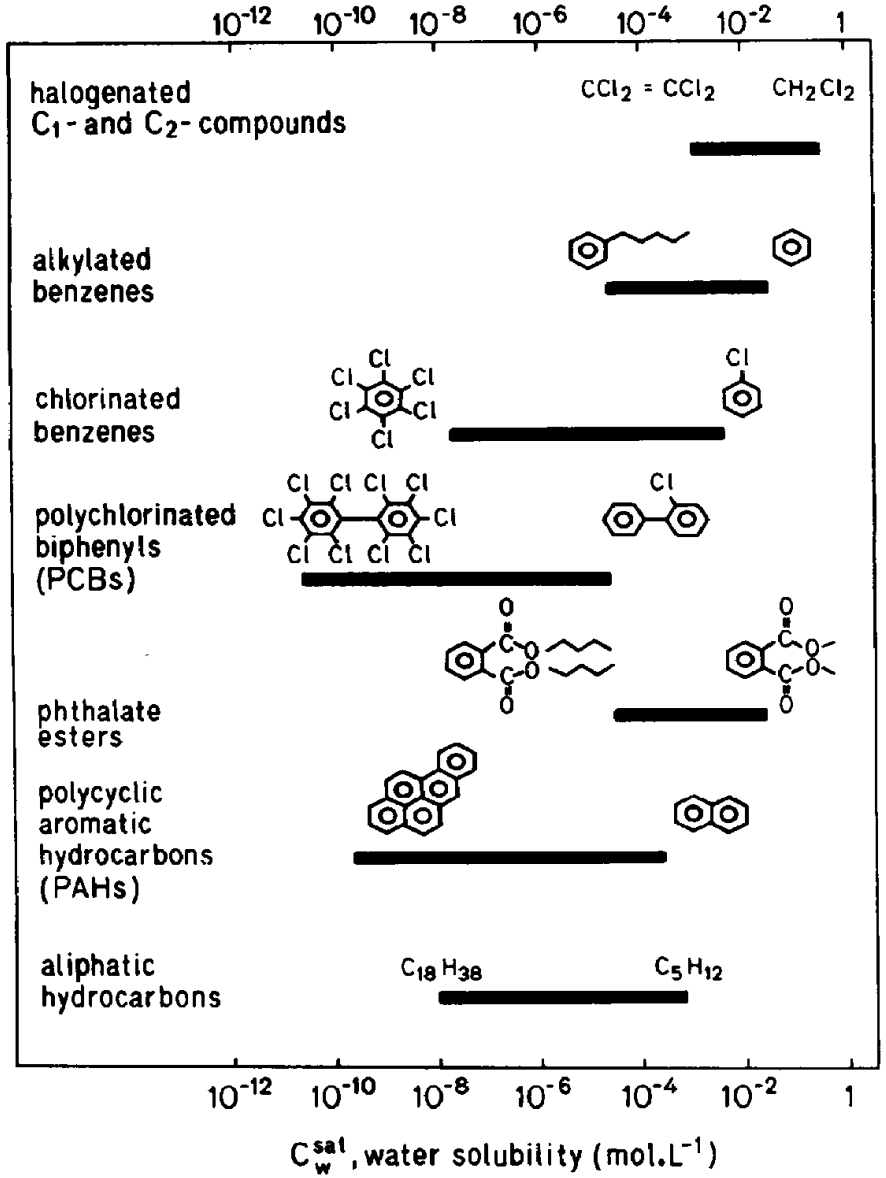


Sources for K_{ow} :

- Data bases (Beilstein)
- USEPA: Winsuite
- Wikipedia
- Chemdraw
- Accelrys Draw

Schwarzenbach (1993)

Water solubility – compound classes



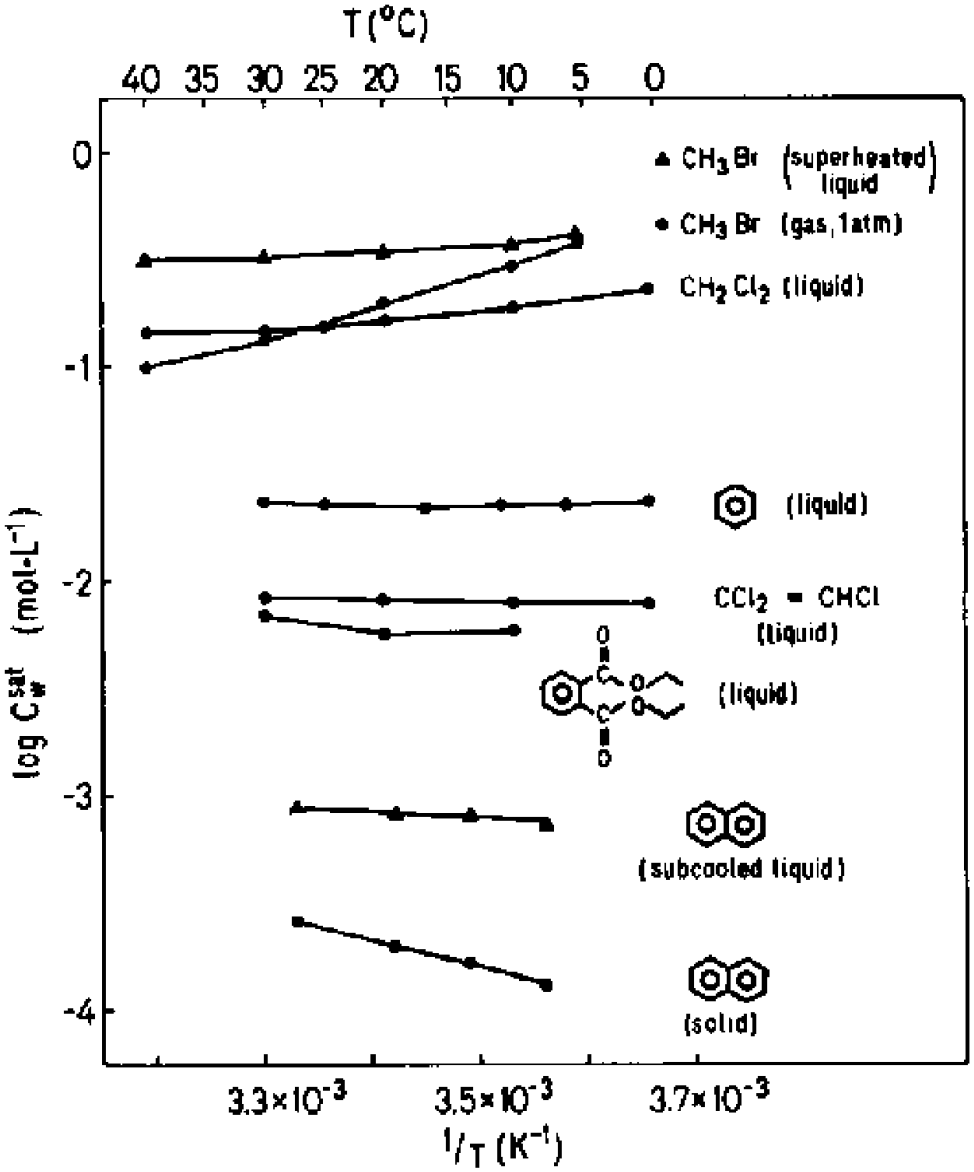
Schwarzenbach (1993)

Water Solubility

The solubility in water (saturation concentration) depends upon :

- **Molecular structure**
  (size, hydrophobicity of the surface)
- **Physico chemical parameters**
  (pressure, temperature)
- **Water chemistry**
  (ionic strength, colloidal and dissolved organic compounds)
- **Acid/basic character**
  (water pH)

Water solubility vs. Temperature



Schwarzenbach (1993)

Water solubility - K_{OW}

$$\log K_{OW} = -a \log C_W^{sat} + b$$

K_{OW} = Octanol-water partition coefficient [-]

a, b = Empirical factors [-]

C_W^{sat} = Solubility in water

Water solubility - K_{OW}

Parameters for the calculation of water solubility for different compound classes from logK_{OW} :

mol/l

			$\log K_{ow} = -a \log C_w^{sat}(1, L) + b$	
Set of Compounds	n	R^2	$a(\pm\sigma)$	$b(\pm\sigma)$
Alkanes	16	0.91	0.81	−0.20
Polycyclic aromatic hydrocarbons	8	0.99	0.87(±0.03)	0.68(±0.16)
Substituted benzenes				
Only nonpolar substituents	23	0.98	0.86(±0.03)	0.75(±0.09)
Including polar substituents	32	0.86	0.72(±0.05)	1.18(±0.16)
Phthalates	5	1.00	1.06(±0.03)	−0.22(±0.09)
PCBs	14	0.92	0.85(±0.07)	0.78(±0.47)
Alcohols	41	0.94	0.90	0.83
Miscellaneous pesticides	14	0.81	0.84(±0.12)	0.12(±0.49)

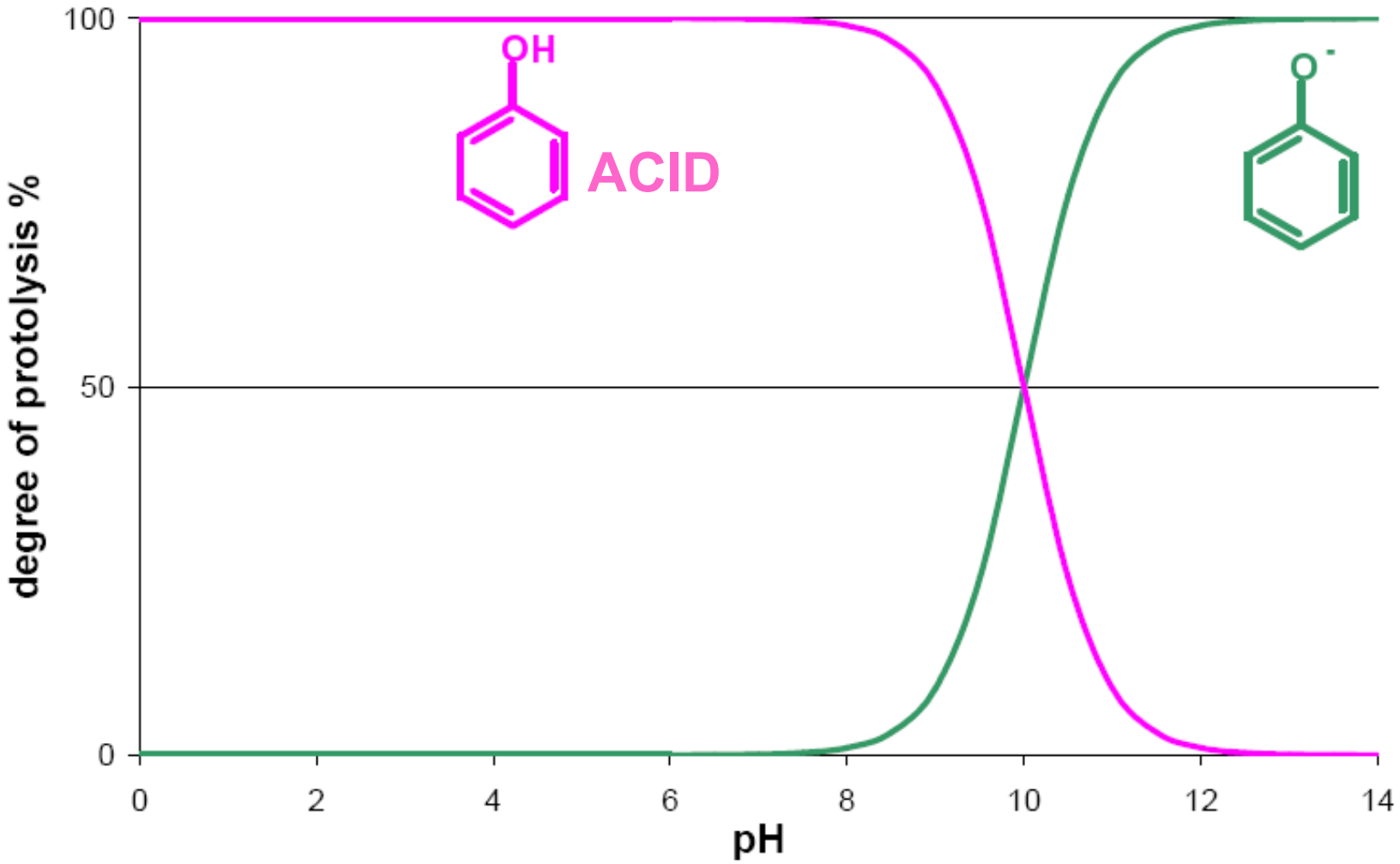
n : number of observations

σ : standard deviation

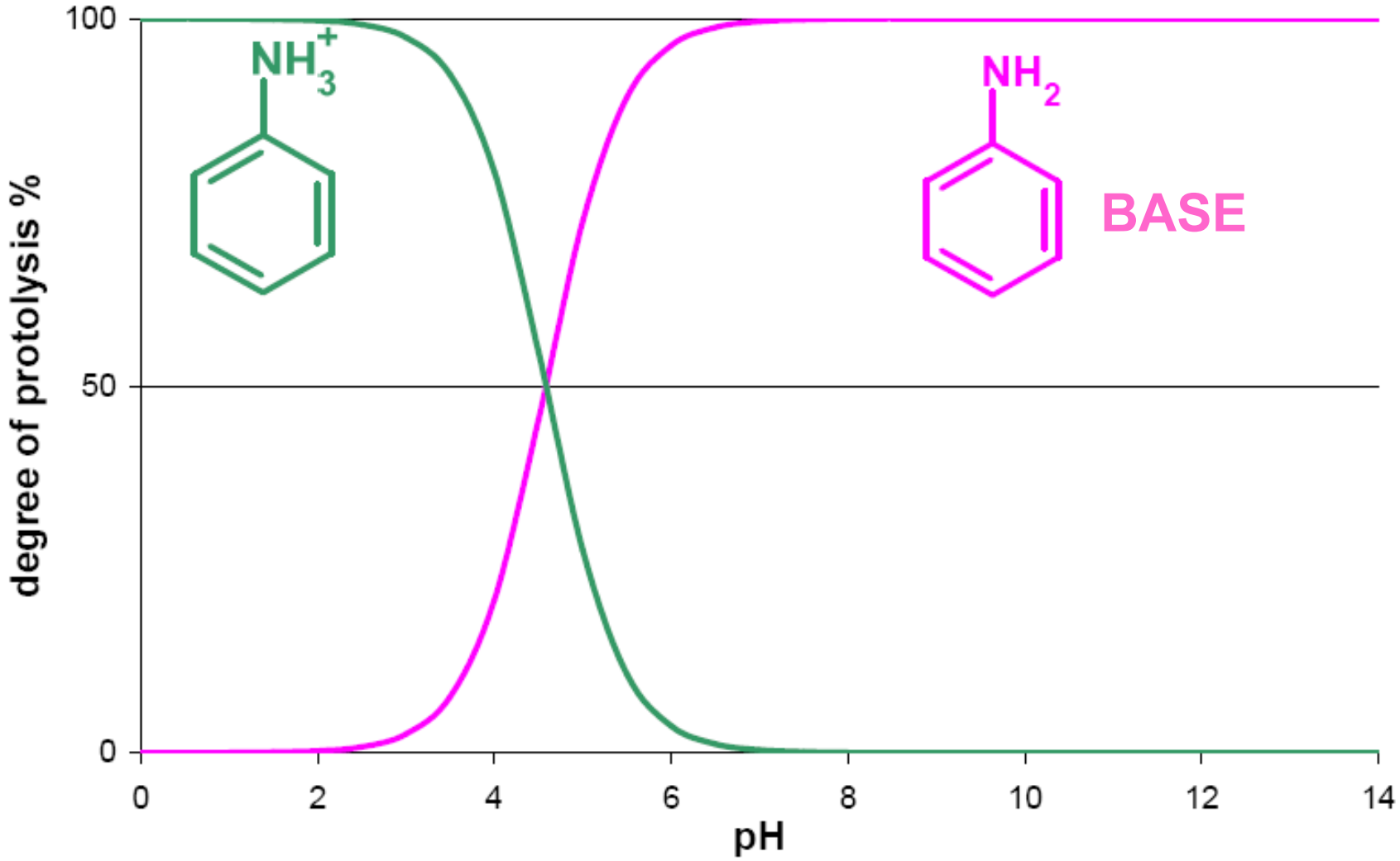
R^2 : correlation coefficient of the partial regression line

Schwarzenbach et al. (1993)

Acid-Base Properties



Acid-Base Properties



Acid-Base Properties

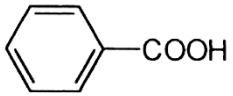
For natural systems:

- **Concentration of organic compounds is commonly very low**
- **pK_a/pK_b of organic compounds suggest them to be weak to very weak**

This means:

- **pH of the water is rarely determined by organic compounds**
- **BUT: water pH determines the speciation of organic molecules (speciation= relative proportion of uncharged to charged)**

Acid-Base Properties: Influence of structure

Name	Formula	pK_a	Name	Formula	pK_a
Acetic acid	CH_3COOH	**4.76**	Trimethyl acetic acid	$(CH_3)_3CCOOH$	**5.05**
Acrylic acid	$CH_2=CHCOOH$	**4.26**	Isobutyric acid	$(CH_3)_2CHCOOH$	**4.85**
Chloroacetic acid	$ClCH_2COOH$	**2.81**	Propionic acid	CH_3CH_2COOH	**4.88**
Dichloroacetic acid	$Cl_2CHCOOH$	**1.30**	Acetic acid	CH_3COOH	**4.76**
Trichloroacetic acid	Cl_3CCOOH	**0.65**	**Formic acid**	$HCOOH$	**3.77**
β-Chloro-propionic acid	$ClCH_2CH_2COOH$	**4.1**	Trifluoro-acetic acid	F_3CCOOH	**0.23**
α-Chloro-propionic acid	$CH_3CHClCOOH$	**2.8**	Benzoic acid		**4.22**

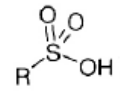
	pK_B	Name	Formula	pKa	
	3.29	Dimethylamin	$(CH_3)_2NH$	**10.71**	
	3.32	tert. Butylamin	$(CH_3)_3CNH_2$	**10.68**	
Increasing Alkalinity	**3.36**	Methylamin	CH_3NH_2	**10.64**	***Decreasing Alkalinity***
	4.26	Trimethylamin	$(CH_3)_3N$	**9.74**	
	4.64	Benzylamin	$C_6H_5CH_2NH_2$	**9.36**	
	4.75	**Ammonial**	**NH_3**	**9.25**	
	9.42	Anilin	$C_6H_5NH_2$	**4.58**	

pKa is valid for the reaction: $R^1R^2R^3NH^+ \rightleftharpoons R^1R^2R^3N + H^+)$

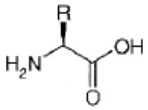
Acid-Base properties - substructures

pK_a Values of Acids

sulfonic acids
pK_a 1



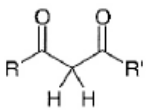
amino acids
pK_a 2-2,5



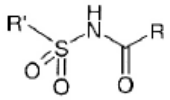
carbonic acids
pK_a 4-5



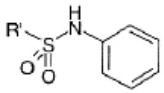
1,3 diketones (in a ring like vinylogous carbonic acids)
pK_a 4-5



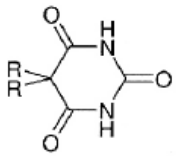
sulfonamides
pK_a 5



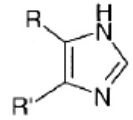
pK_a 6-8



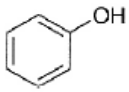
5,5 disubstituated barbiturates
pK_a 7-8



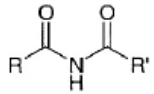
annulated imidazoles
pK_a 9



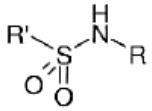
phenols
pK_a 10



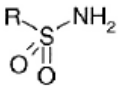
imides
pK_a 10



sulfonamides
pK_a 10



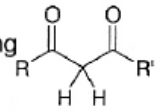
pK_a 10



thio-alcohols
pK_a 10



1,3 diketones (aliphatic,depending on R in chain)
pK_a 11-14

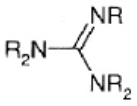


alcohols
pK_a 17

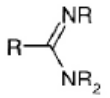


pK_b Values of Bases

guanidines
aliphatic pK_b 0,5-1
aromatic pK_b 3-5



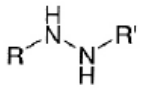
amidines
aliphatic pK_b 2-3
aromatic pK_b 5-10



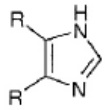
amines
aliphatic pK_b 4-5
sterically hindered
pK_b 5-6



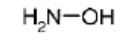
hydrazines
pK_b 7



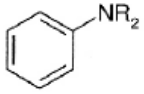
imidazoles
pK_b 7



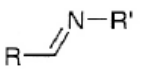
hydroxylamine
pK_b 8



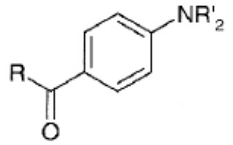
aromatic amines
pK_b 9-10



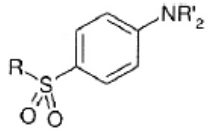
imines, also pyridine, quinoline
pK_b 9-10



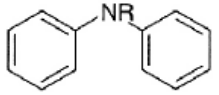
phenylogous amides
pK_b 11-12



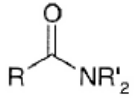
phenylogous sulfonamides
pK_b 12



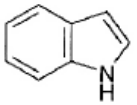
pK_b > 14



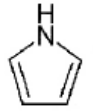
acid amides
aliphatic pK_b 14
aromatic pK_b 16



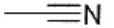
indole
pK_b 16



pyrrole
pK_b 16

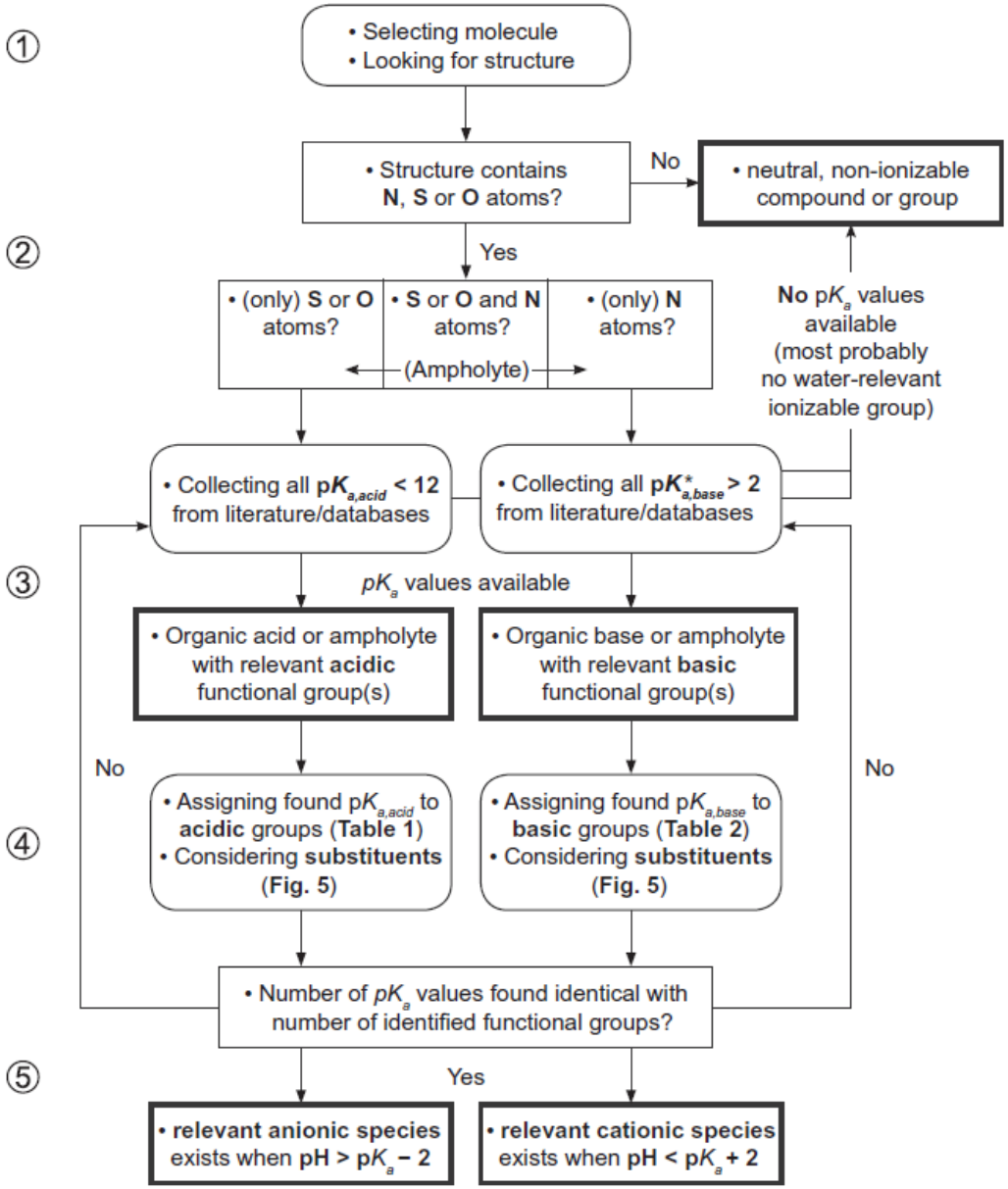


nitriles
pK_b >20



sulfonamides
imides
pK_b >20

Charge – pH: flowchart



Acid-Base Properties - substructures

For more sub-structures and flow chart use consult:

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journal homepage: www.elsevier.com/locate/chemosphere



Review

A guideline for the identification of environmentally relevant, ionizable organic molecule species

Mario Schaffer, Tobias Licha *

Geoscience Centre, Dept. Applied Geology, University of Göttingen, Goldschmidtstr. 3, 37077 Göttingen, Germany

HIGHLIGHTS

- Guideline for the identification of ionizable functional groups in organic compounds.
- Definition of important terms and instruction for the correct assignment of pK_a values.
- Systematic tables including >30 organic compound classes and their typical pK_a ranges.
- Comprehensive application of the method on the example of environmentally relevant substances.

Extending logK_{ow} to partially ionic compounds: logD

Important if pKa is less than 2 units away from the respective water pH

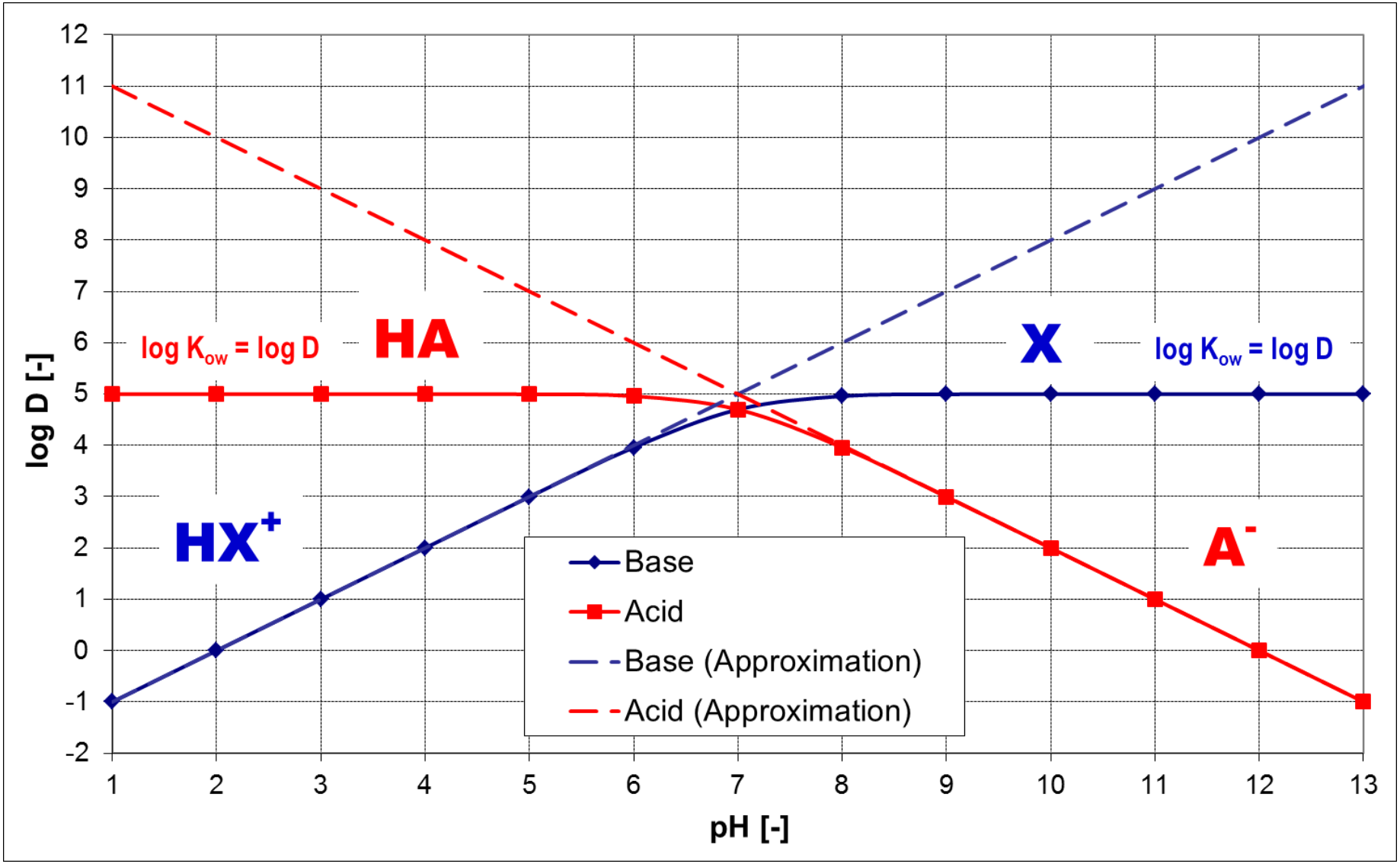
For acids (negative charge) :

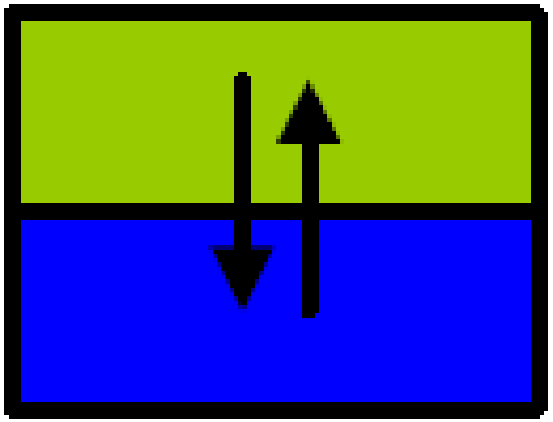
$$\log D = \log K_{OW} - \log\left(1 + 10^{(pH - pK_a)}\right)$$

For bases (positive charge) :

$$\log D = \log K_{OW} - \log\left(1 + 10^{(pK_a - pH)}\right)$$

Extending logK_{ow} to partially ionic compounds: logD

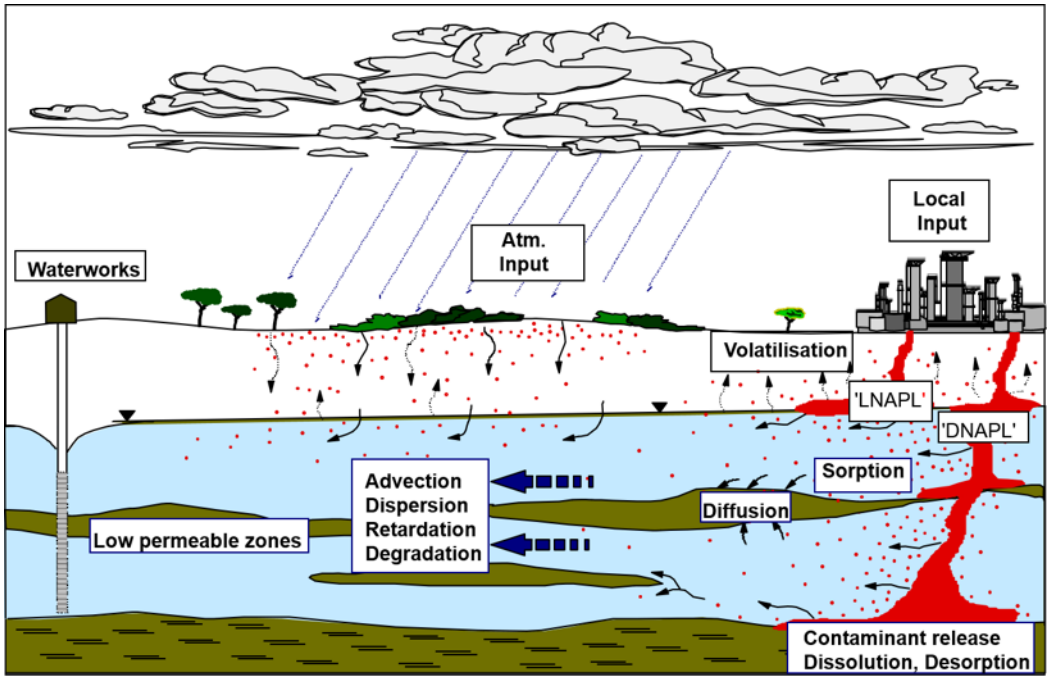




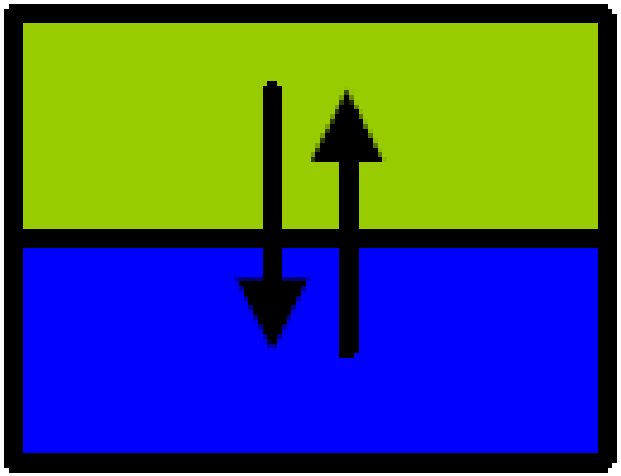
Distribution Equilibrium:

water/soil (aquifer material)

- Contaminants are transported with the groundwater flow direction
- Light nonaqueous phase liquids (LNAPLs) accumulate above groundwater table
- Dense nonaqueous phase liquids (DNAPLs) sink downward and accumulate at non permeable zones



Partitioning coefficients - soil/water



Several partitioning coefficients are known

Underlying process is called *sorption*

- several concepts are used to describe / quantify the soil/water partitioning of organic compounds
- partitioning is driven by a wide range of specific and non-specific interactions
- concentration dependencies of partitioning largely depend on interaction types

Processes involved in soil/water partitioning: Sorption

1. Mechanistical approach

Definition of Sorption

Sorption = *interfacial process, a distinction is drawn between:*

Adsorption	Enrichment of substances at the **interface** between solid, gaseous or liquid phase.
Absorption	Process of **transferring** ions, atoms or molecules across an interface into another phase.
Desorption	The reversible process of adsorption and absorption

***ad**sorption,*

***ab**sorption und*

***de**sorption.*

Which process is more relevant in subsurface transport of organics?

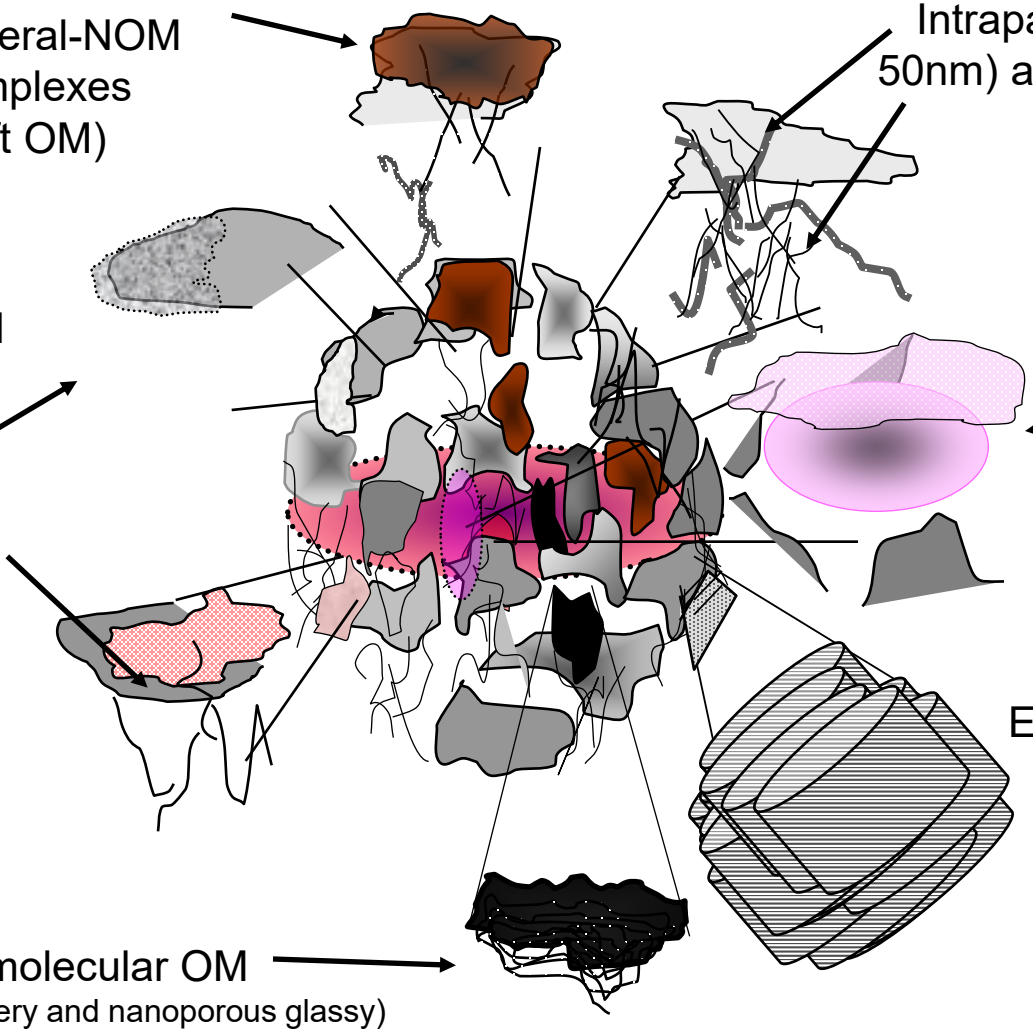


ABsorption



ADsorption

Domains of sorption/heterogeneity



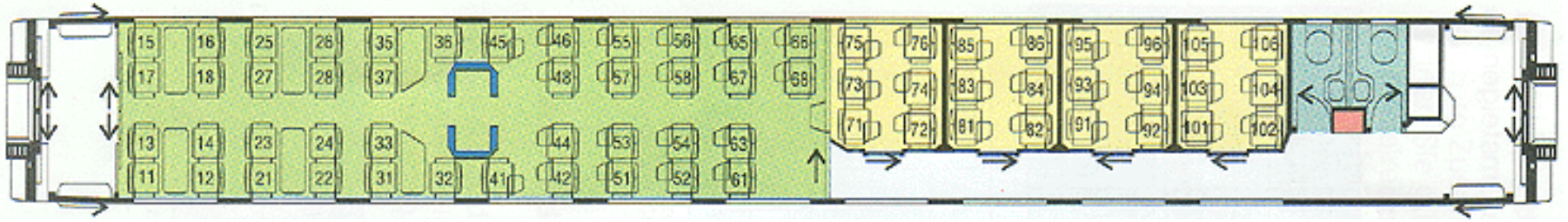
Main characteristics

Capacity of adsorption:

Is it a single layer or multilayer adsorption process?

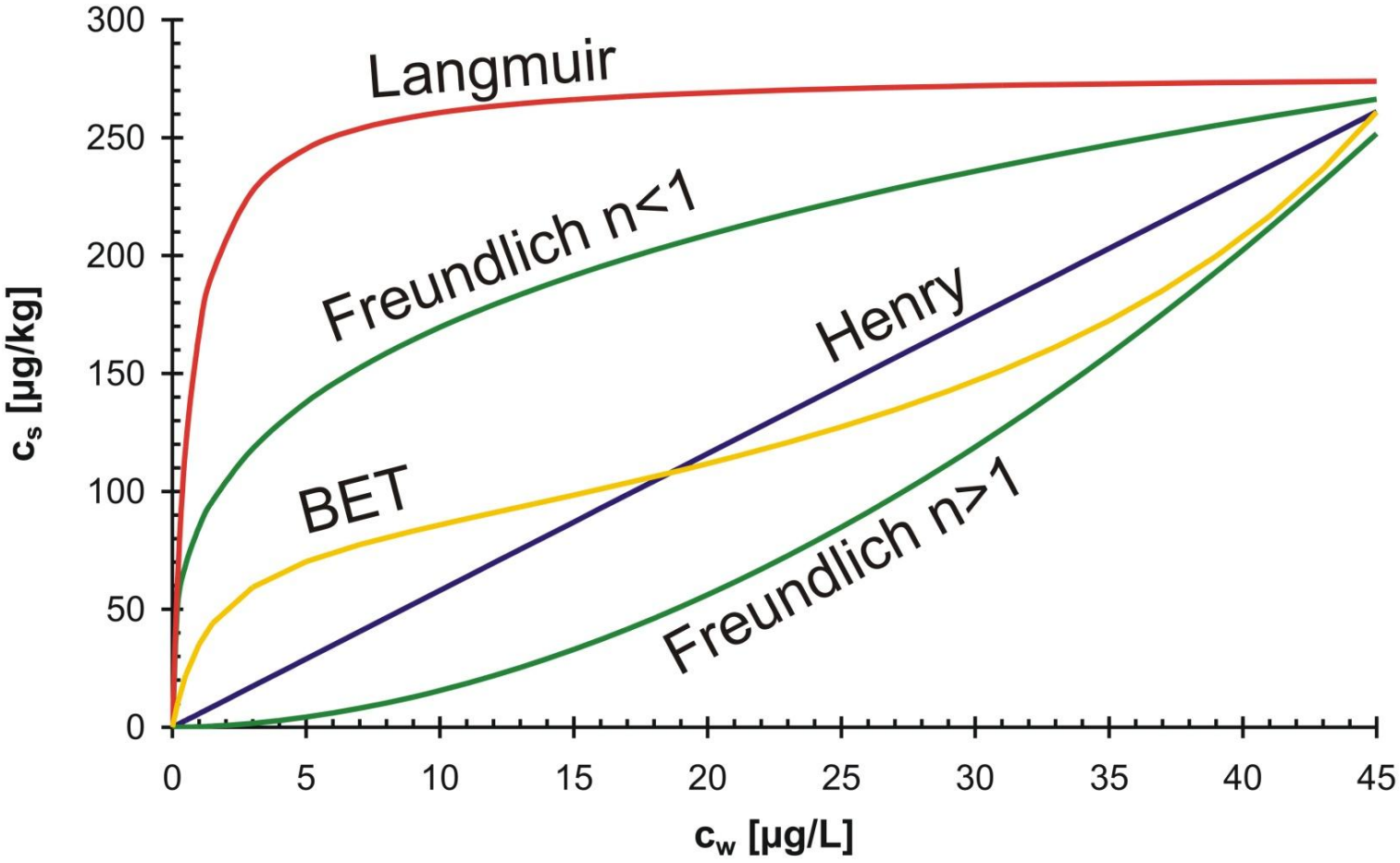






Isotherms

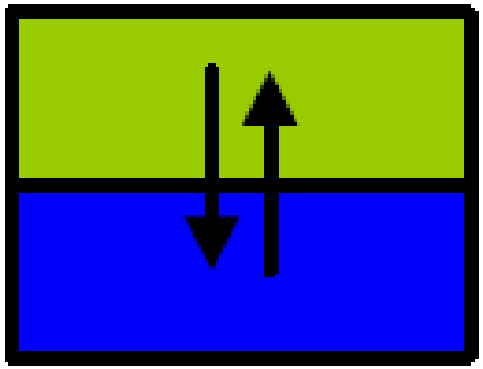
Isotherms are empiric relations. They describe the partition of a substance between different phases at constant temperature.



Henry-Isotherm, Definition K_d

soil, aquifer material

water



K_d

$$K_d = \frac{c_s}{c_w}$$

K_d = Soil-water distribution coefficient (L kg^{-1})

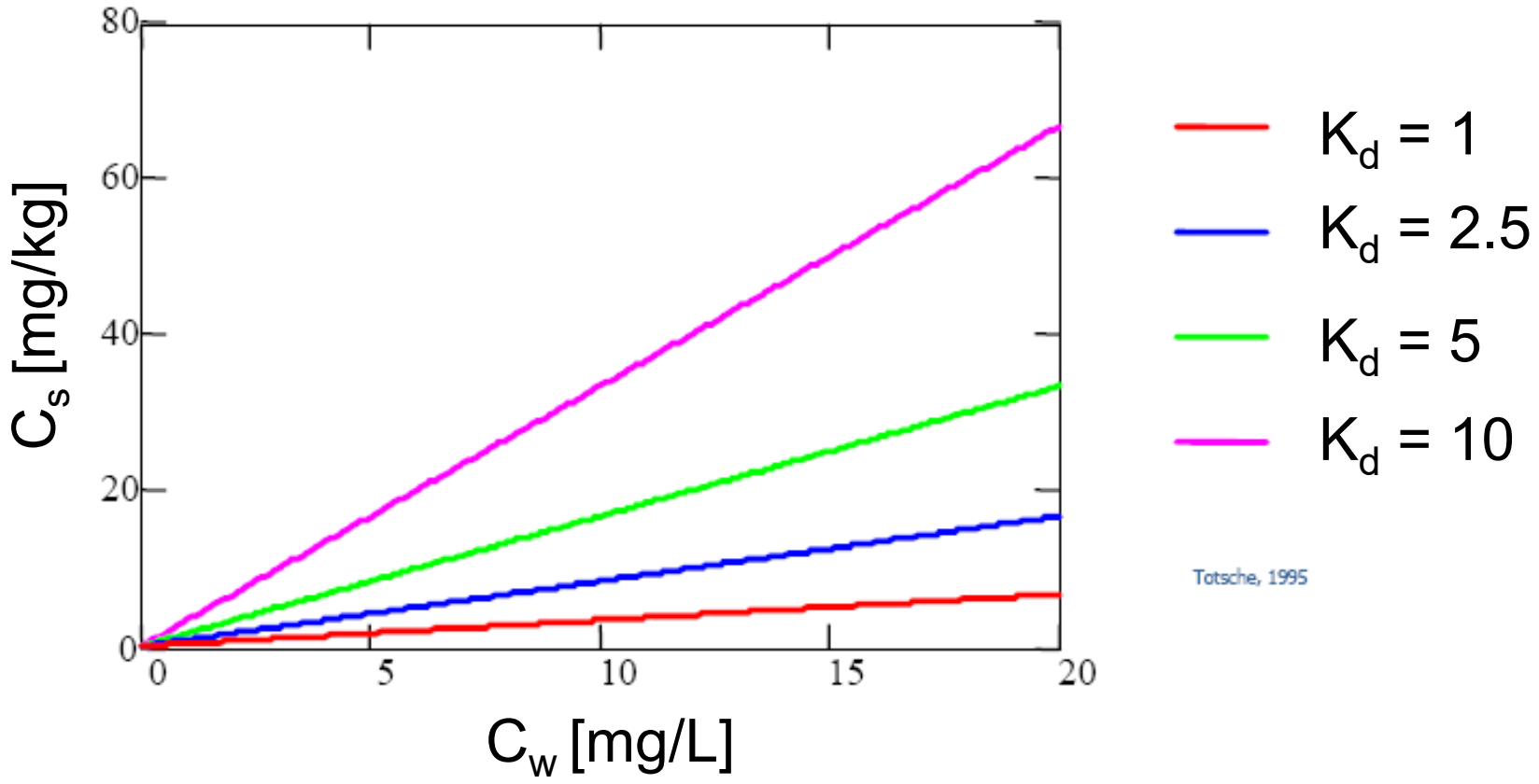
C_S = Concentration of substance on the solid phase (µg kg^{-1})

C_W = Concentration of substance in water (µg L^{-1})

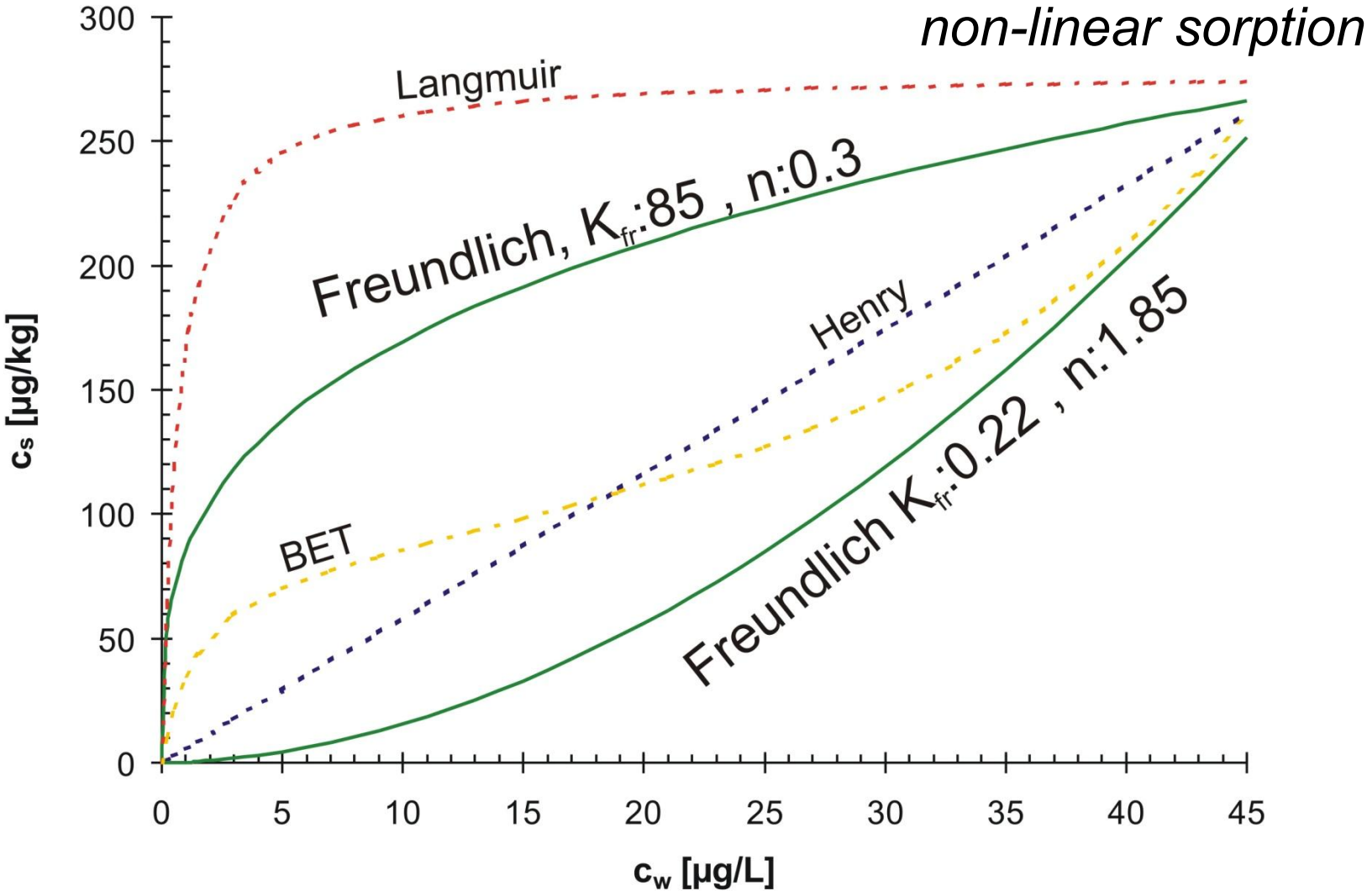
Linear Isotherms (Henry)

Partitioning:
linear sorption

$$c_s = K_d\, c_w$$



Freundlich - Isotherm



Freundlich - Isotherm

$$c_s = K_{fr}\, c_w^{\,n}$$

K_{fr} = Freundlich distribution coefficient $(L\ kg^{-1})^{(1-n)}$

n = Freundlich exponent [-]

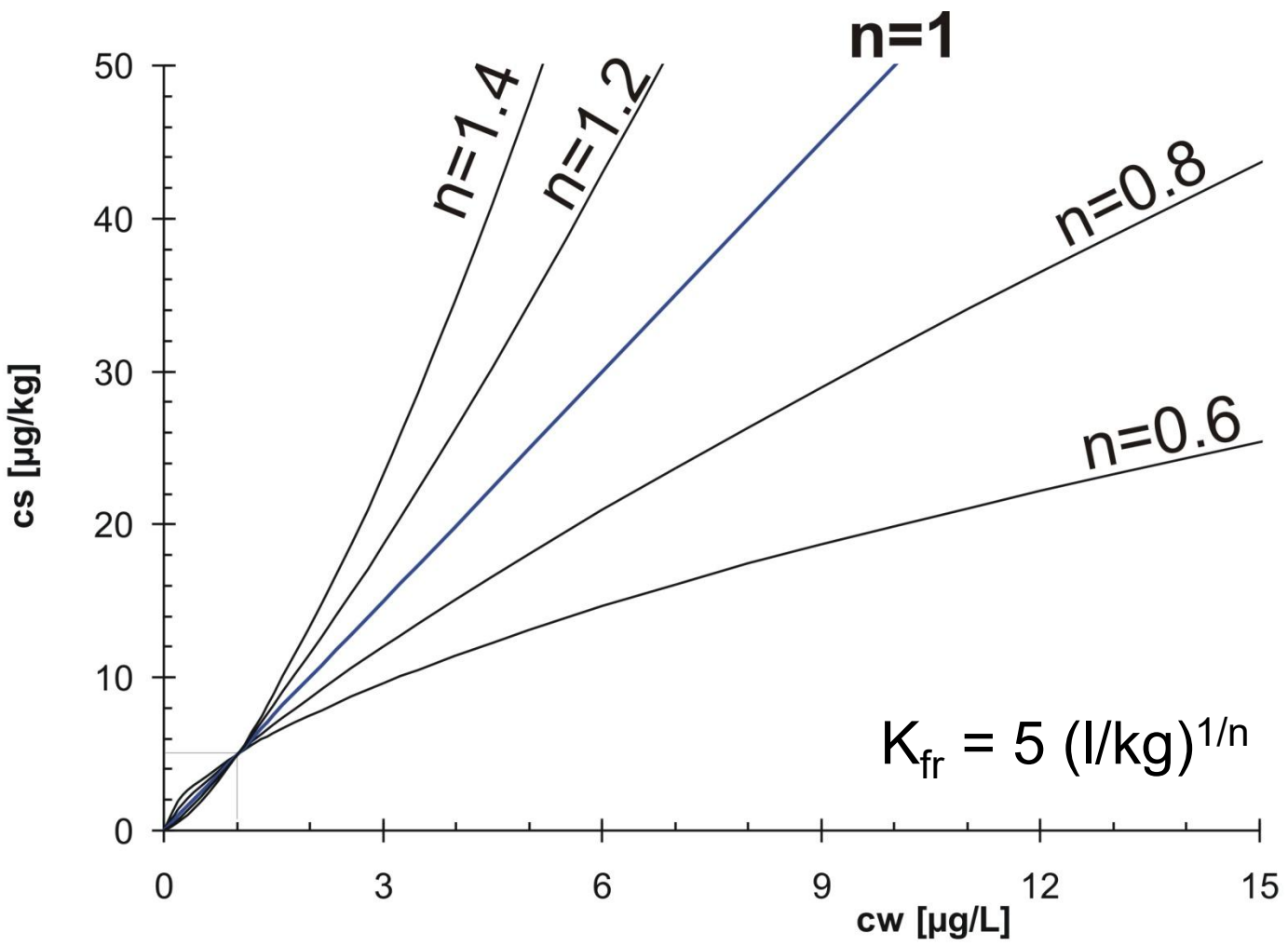
C_S = Concentration of substance on the solid phase ($\mu g\ kg^{-1}$)

C_W = Concentration of substance in water ($\mu g\ L^{-1}$)

also found as: $c_s = K_{fr}\, c_w^{\frac{1}{n}}$ $\qquad \Gamma_2^{\,v} = K_{fr}\, c_2^{eq\,\frac{1}{n}}$

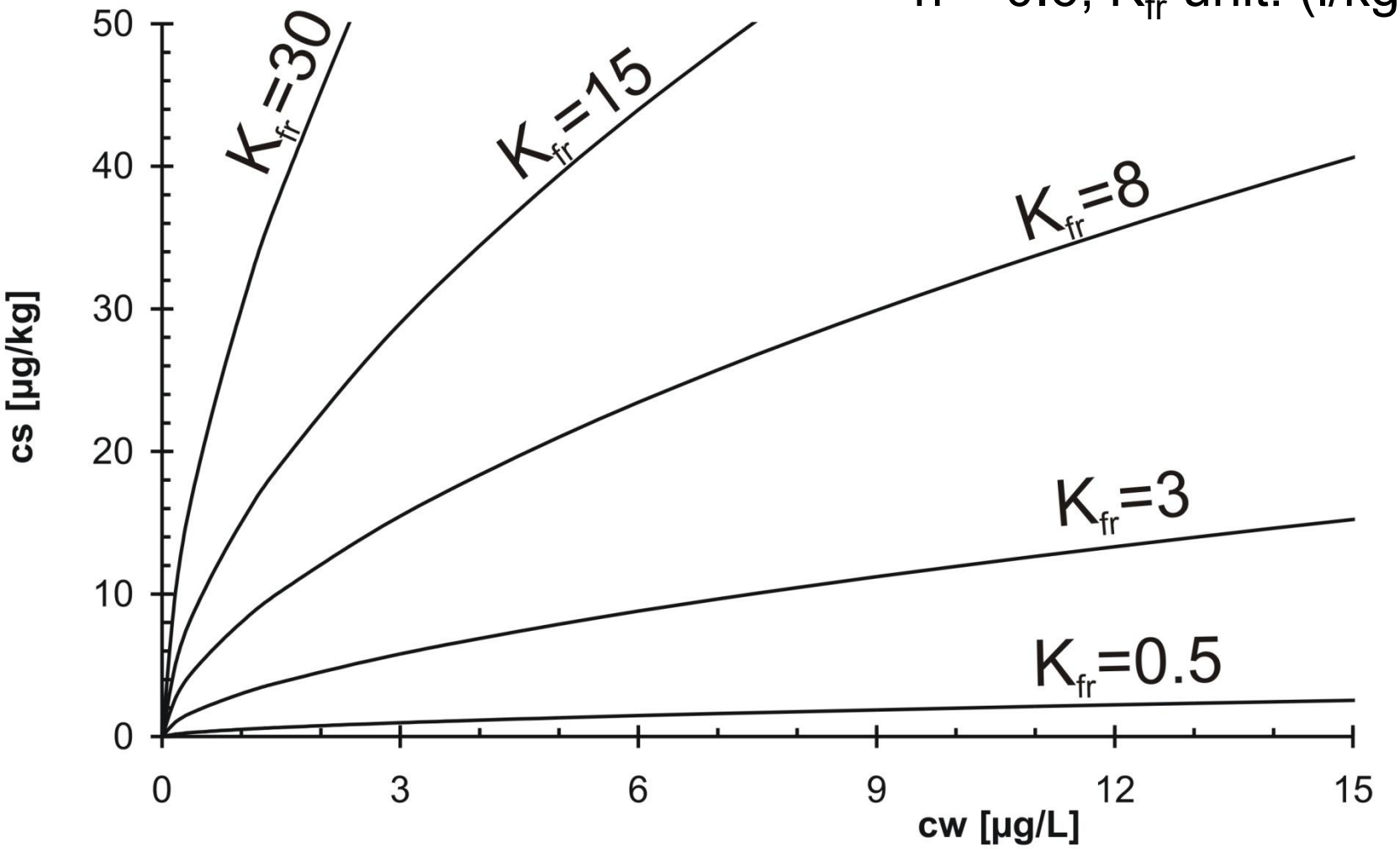
☞ unit of K_{fr} tells you which form you are dealing with

Freundlich - Isotherm $c_s = K_{fr}\, c_w^{\,n}$



Freundlich - Isotherm $c_s = K_{fr}\, c_w^{\,n}$

n = 0.6, K_{fr} unit: $(l/kg)^{1/n}$



Introduction to concepts for estimating/predicting sorption without determining isoterms by lab experiments

The K_{oc} concept

K_{OC}

Analogue to K_{OM} the following equation is valid for K_{OC}:

$$K_{OC} = \frac{C_{OC}}{C_W} = \frac{K_d}{f_{OC}}$$

K_{OC} = partition coefficient organic carbon – water (L kg^{-1})
K_d = partition coefficient soil – water (L kg^{-1})
C_{OC} = concentration of contaminant related to organic carbon (-)
C_W = concentration of contaminant in water (kg L^{-1})
f_{OC} = content of organic carbon (-)

f_{OC}

Because the mass of OC is about 50 % of the OM, the following empirical formula can be used:

$$f_{OM} \sim 2\, f_{OC}$$

f_{OM} = content of organic matter (-)
f_{OC} = content of organic carbon (-)

Component	Mole Ratio C	H	N	O	Molecular Mass average (u) [a]	% Aromaticity
Biogenic Molecules						
Proteins	1.0	1.6	0.4	0.2		< 10
Collagen (protein)	1.0	1.7	0.19	0.31		< 10
Cellulose (polysaccharide)	1.0	1.7	< 0.01	0.84	*ca.* 10^6 (cotton)	< 10
Chitin (polysaccharide)	1.0	1.8	0.13	0.64		< 10
Lignin (alkaline extract)	1.0	1.1	< 0.01	0.40		28
Lignin (org. solvent extract)	1.0	0.98	< 0.01	0.33		34
Diagenetic Materials						
Fulvic acids						
soil leachate	1.0	1.04	< 0.1	0.53		36
brown lake water	1.0	0.88	< 0.1	0.55		35
river water	1.0	1.62	< 0.1	1.09		29
groundwater	1.0	1.04	< 0.1	0.51		24
Suwannee River fulvic acid	1.0	0.87	0.1	0.53	2000	25
Humic acids						
brown lake water	1.0	0.80	< 0.1	0.54		40
river water	1.0	1.48	< 0.1	0.91		38
"average" soil	1.0	1.15	0.07	0.50	7800	
Aldrich	1.0	0.78	0.01	0.44	9000	41
Suwannee River	1.0	0.94	0.02	0.61	3200	42
Humin	1.0	1.9	0.5	1.1		
Kerogen	1.0	0.4 to 1		0.05 to 0.3		
Combustion-Derived Materials						
NIST diesel soot	1.0	0.1	0.016			~100
BC from Boston Harbor sediment	1.0	1.0	0.07			

Estimation of K_{oc}

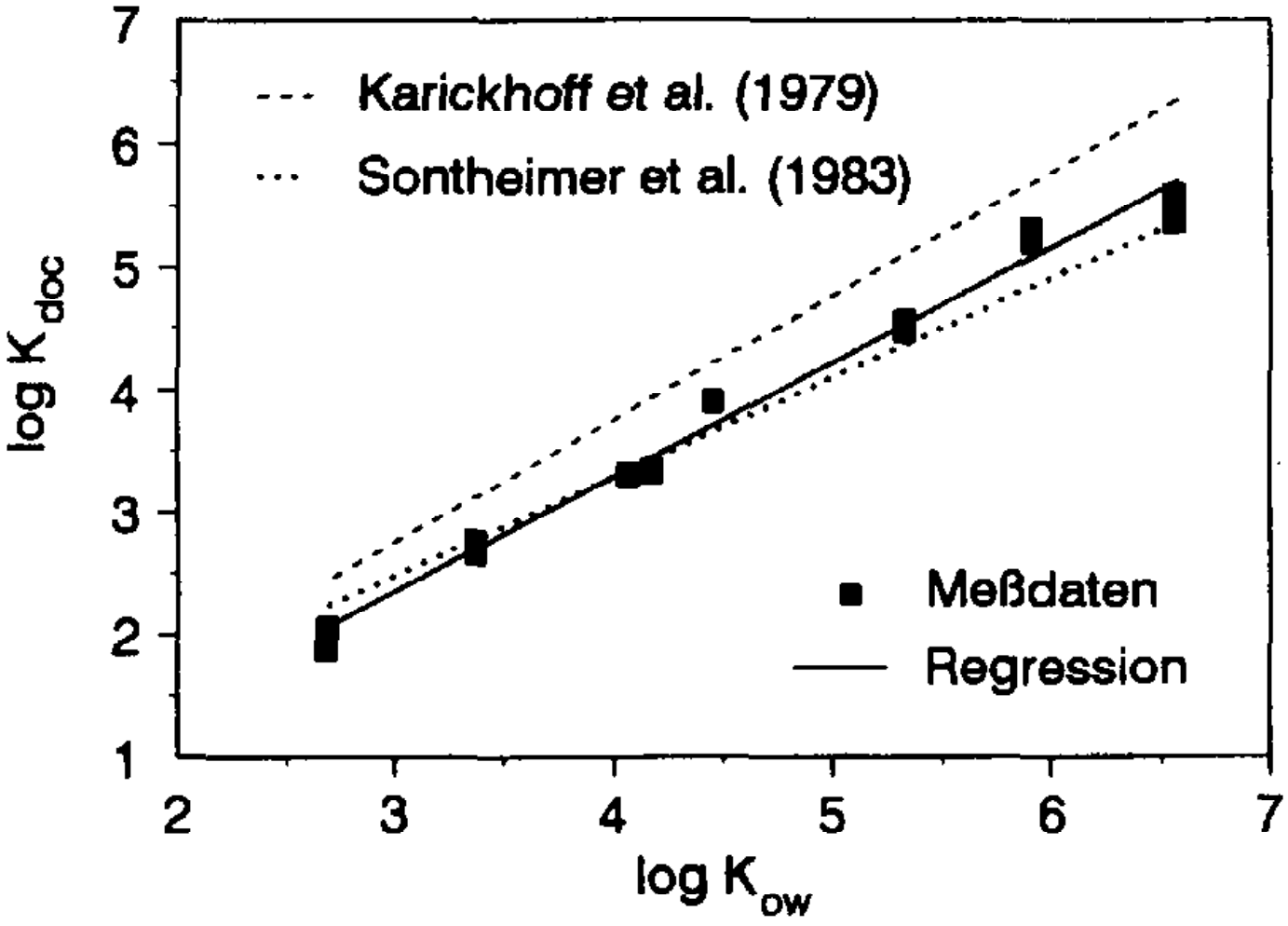
Estimation of K_{OC} by:

- Solubility S_W
- Octanol-water partitioning coefficient K_{OW}

$$\log K_{OC} = a \cdot \log K_{OW} + b$$

$$\log K_{OC} = c \cdot \log S + d$$

Estimation of K_{oc}



Estimation of K_{oc}

There are empiric correlations between K_{OC} and K_{OW} which was investigated by several authors:

universal relations:

$\log K_{OC} = \log K_{OW} - 0.21$	Karickhoff et al. (1979)
$\log K_{OC} = 0.356 \log K_{OW} + 1.15$ $(\log K_{OW} = 1.0 - 2.4)$	Sontheimer et al. (1983)
$\log K_{OC} = 0.807 \log K_{OW} + 0.086$ $(\log K_{OW} = 2.4 - 7.4)$	Sontheimer et al. (1983)

compound specific relations:

$\log K_{OC} = 0.36 \log K_{OW}$	PAH, chlorinated HC	Karickhoff (1984)
$\log K_{OC} = \log K_{OW} - 0.317$	PAH	Means et al. (1980)
$\log K_{OC} = 0.807 \log K_{OW} + 0.068$	aliphatic CHC	Sontheimer et al. (1993)
$\log K_{OC} = 0.904 \log K_{OW} - 0.542$	PAH, benzene derivates	Sontheimer et al. (1993)

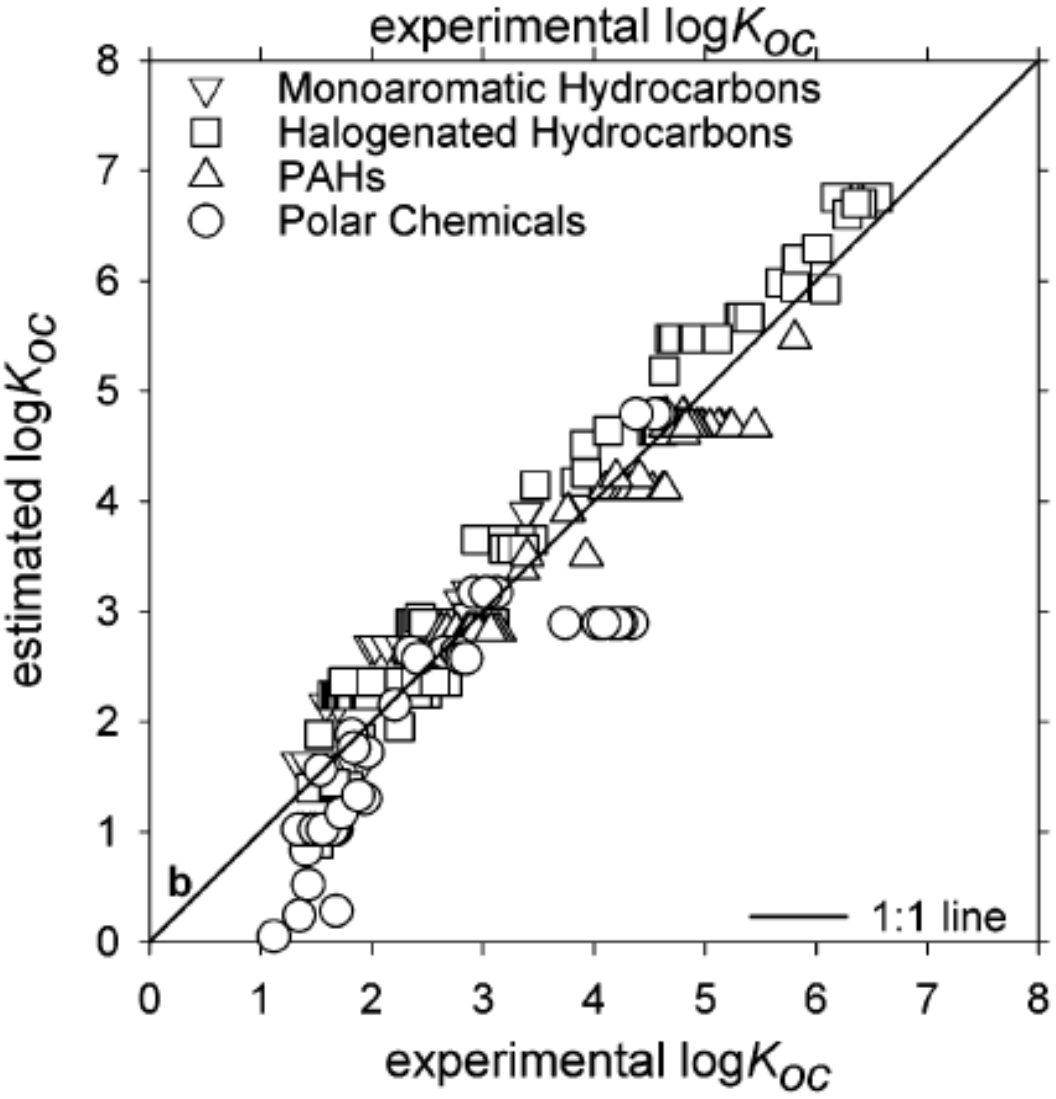
Estimation of K_{oc}

$K_{iOC} = K_{OC}$, $K_{iOW} = K_{OW}$

Set of Compounds	$\log K_{ioc}{}^{b} = a \cdot \log K_{iow} + b$ a	b	$\log K_{iow}$ range [c]	R^2	N [d]
Alkylated and chlorinated benzenes, PCBs (± apolar)	0.74	0.15	2.2 to 7.3	0.96	32
PAHs (monopolar)	0.98	–0.32	2.2 to 6.4	0.98	14
Chlorinated phenols (neutral species; bipolar)	0.89	–0.15	2.2 to 5.3	0.97	10
C_1- and C_2-halocarbons (apolar, monopolar, and bipolar)	0.57	0.66	1.4 to 2.9	0.68!	19
Only chloroalkanes (mix)	0.42	0.93		0.59!	9
Only chloroalkenes (± apolar)	0.96	–0.23		0.97	4
Only compounds including bromine (mix)	0.50	0.81		0.49!	6
All phenylureas (bipolar)	0.49	1.05	0.5 to 4.2	0.62!	52
Only alkylated and halogenated phenylureas, phenyl-methylureas, and phenyl-dimethylurea (bipolar) [e]	0.59	0.78	0.8 to 2.9	0.87	27
Only alkylated and halogenated phenylureas (bipolar) [f]	0.62	0.84	0.8 to 2.8	0.98	13

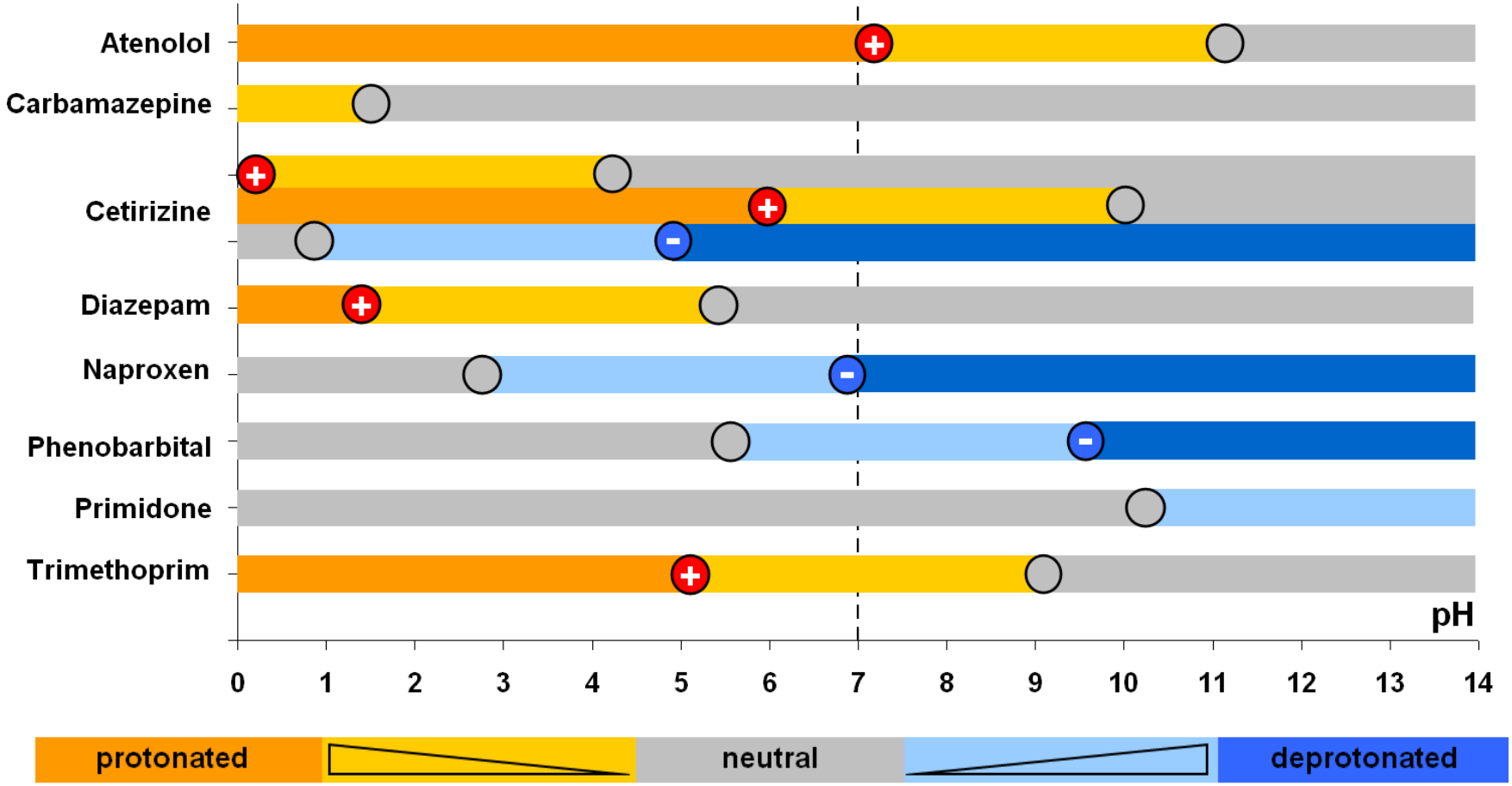
Schwarzenbach, 3rd edition

Limitations of K_{oc} concept



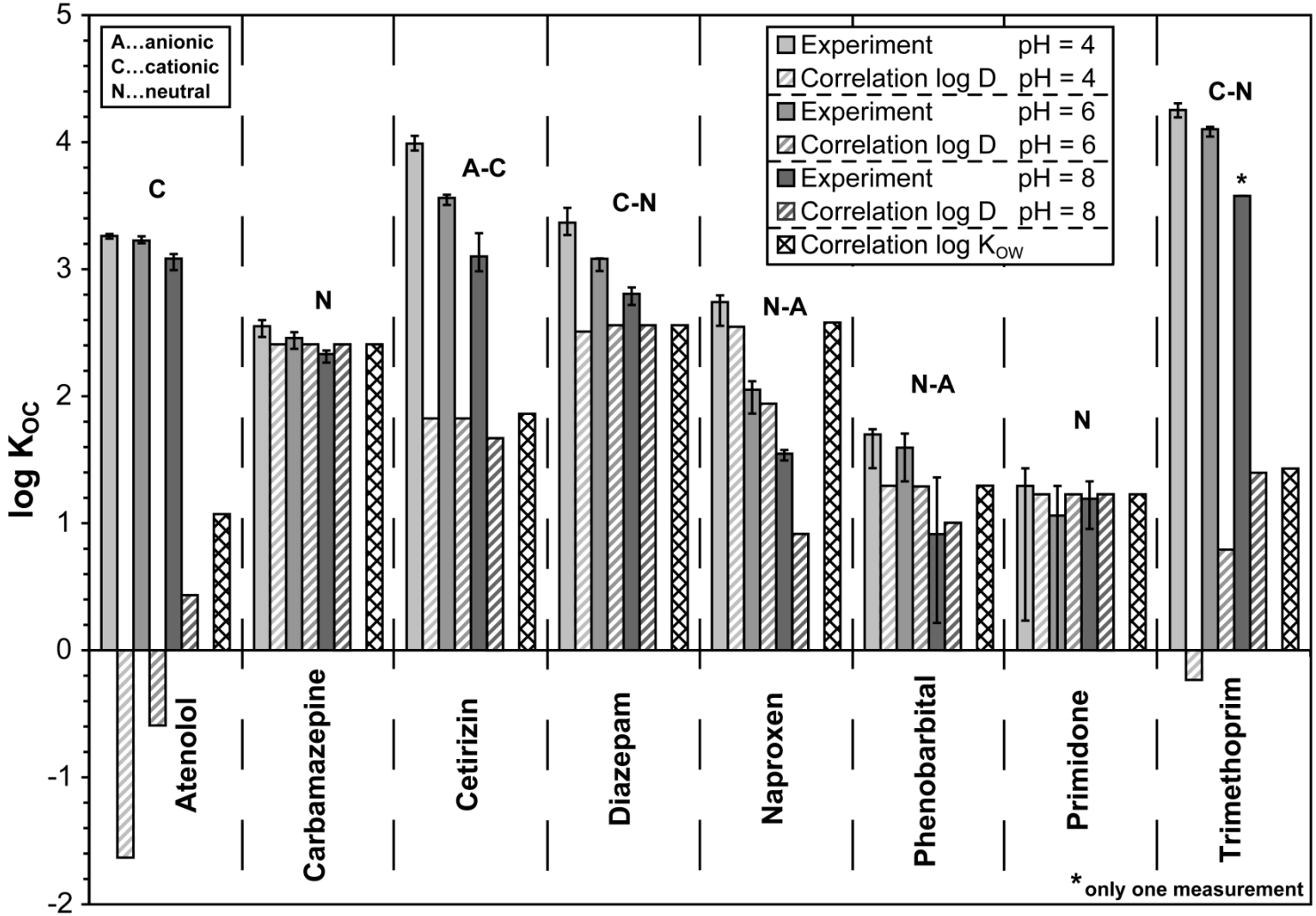
(Nquyen, 2005)

Multispecies-Sorption



Schaffer M, Boxberger N, Börnick H, Licha T, Worch E (2012): Sorption influenced transport of ionizable pharmaceuticals onto a natural sandy aquifer sediment at different pH. Chemosphere 87(5): 513–520.

Multispecies-Sorption: Dependance of pH



Schaffer M, Boxberger N, Börnick H, Licha T, Worch E (2012): Sorption influenced transport of ionizable pharmaceuticals onto a natural sandy aquifer sediment at different pH. Chemosphere 87(5): 513–520.

Limitations of K_{oc} concept

uncharged molecules:

K_{ow} – K_{oc} relations valid, quality of prediction is compound specific

negatively charged molecules (acids):

used logD instead logK_{ow} (see topic II)

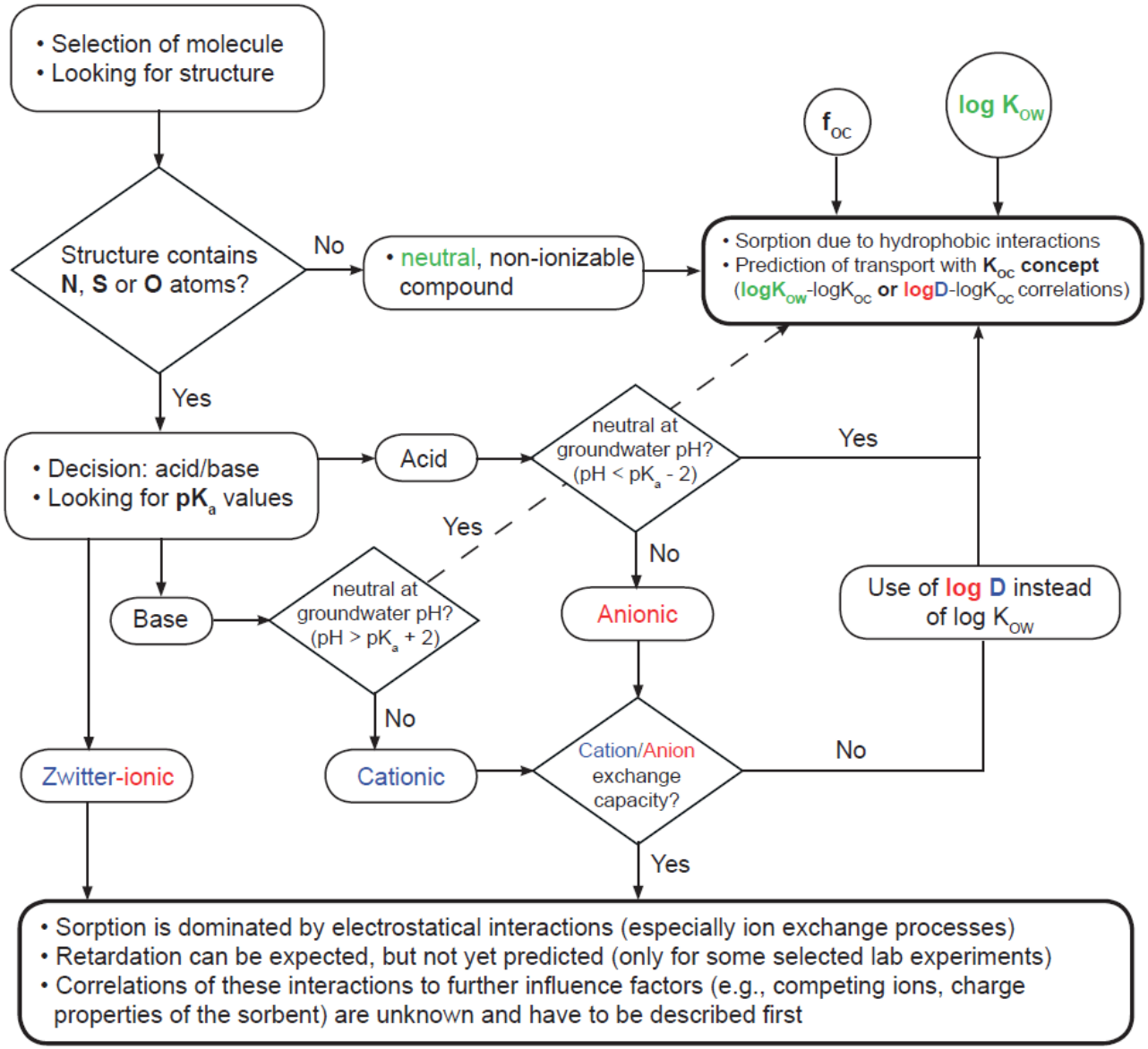
positively charged molecules (bases):

K_{ow}-K_{oc} relations not valid, no prediction possible

zwitterionic molecules (ampholytes):

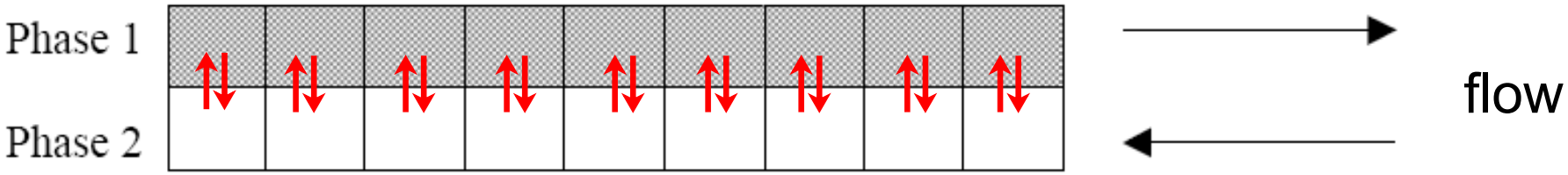
K_{ow}-K_{oc} relations not valid, no prediction possible

Decision tree sorption strength



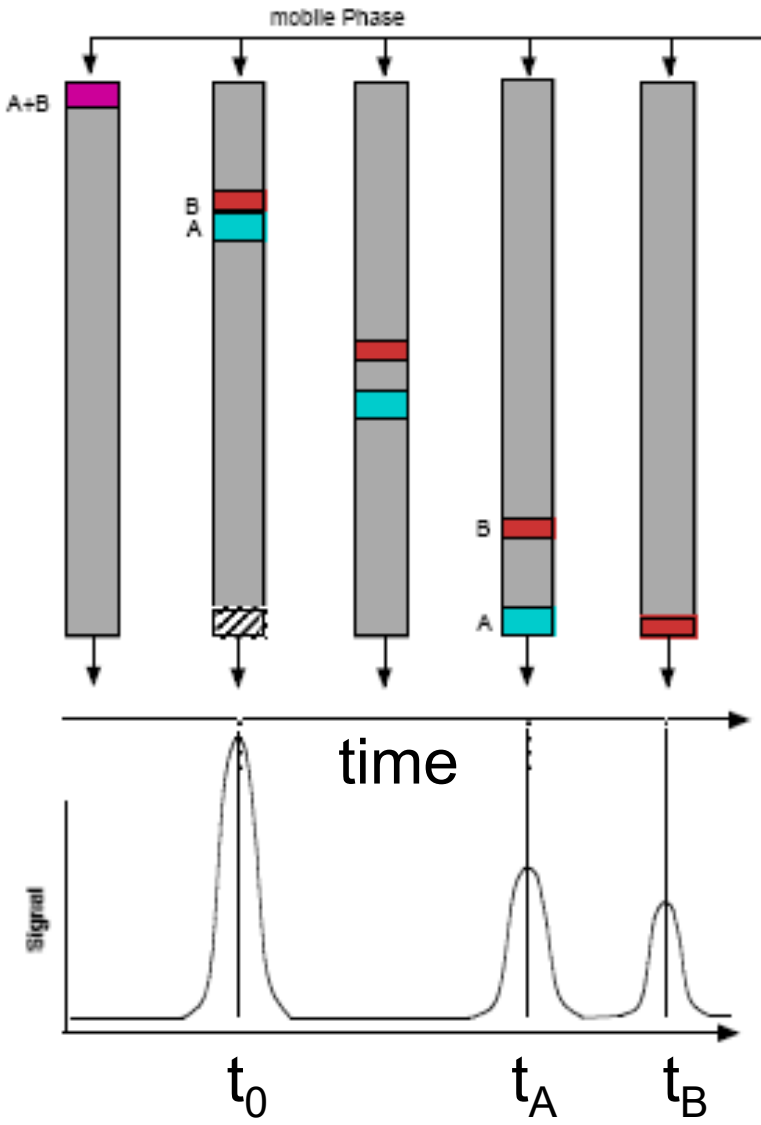
Effect of sorption on advection

Partitioning coefficients - soil/water



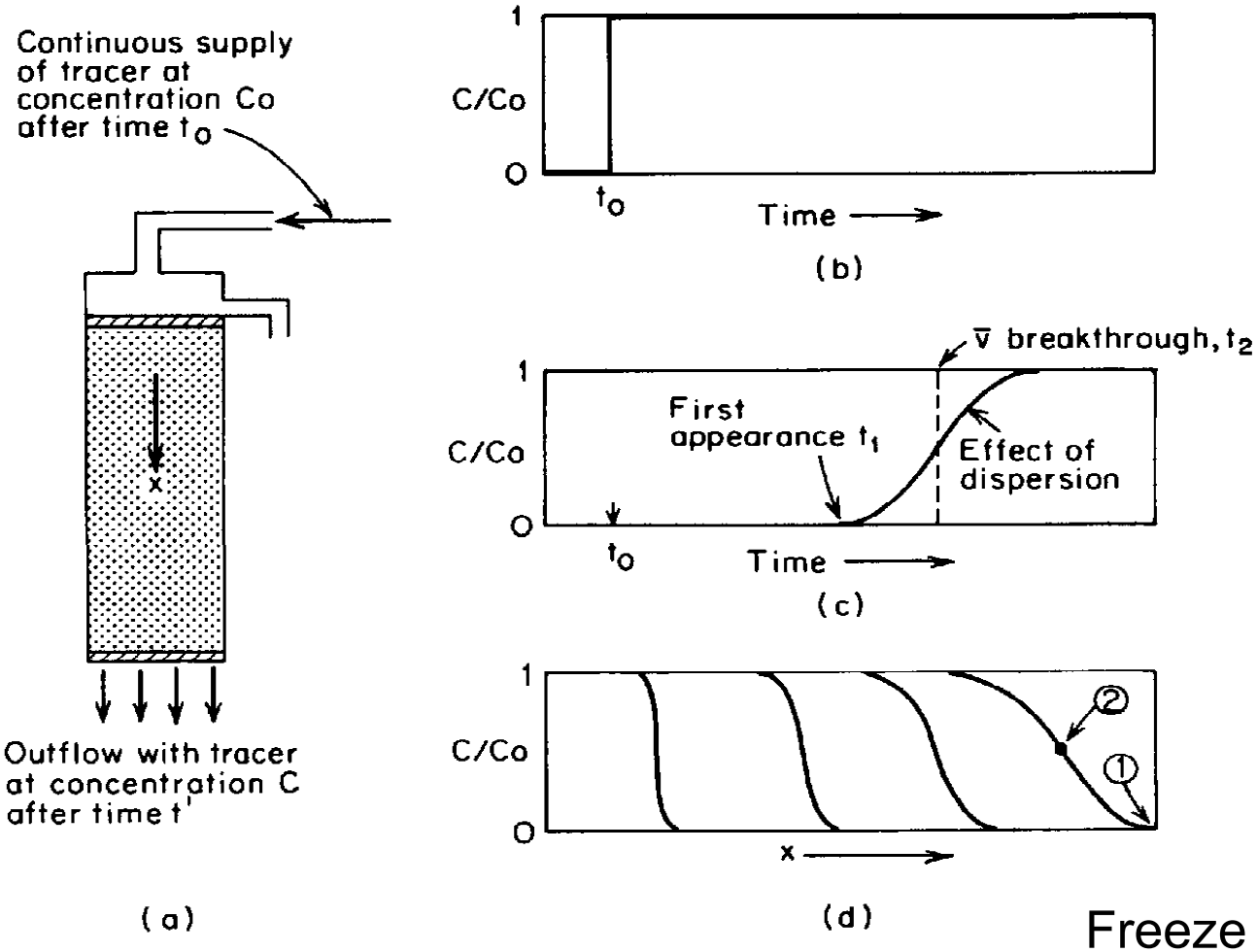
Continuous partition by dynamic flow

Sorption and Retardation



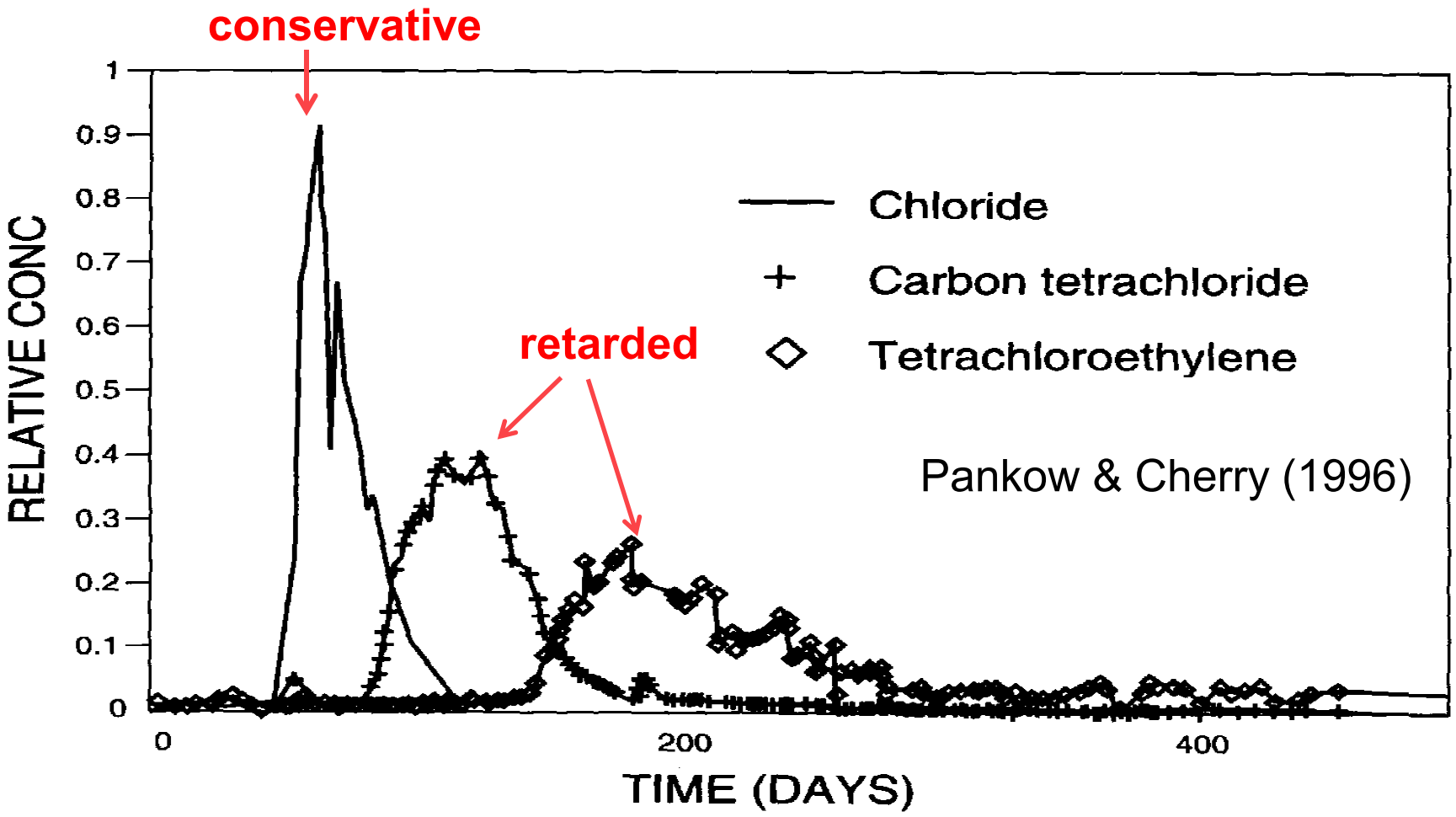
$$R_A = \frac{t_A}{t_0} \qquad R_B = \frac{t_B}{t_0}$$

Retardation



Freeze & Cherry (1979)

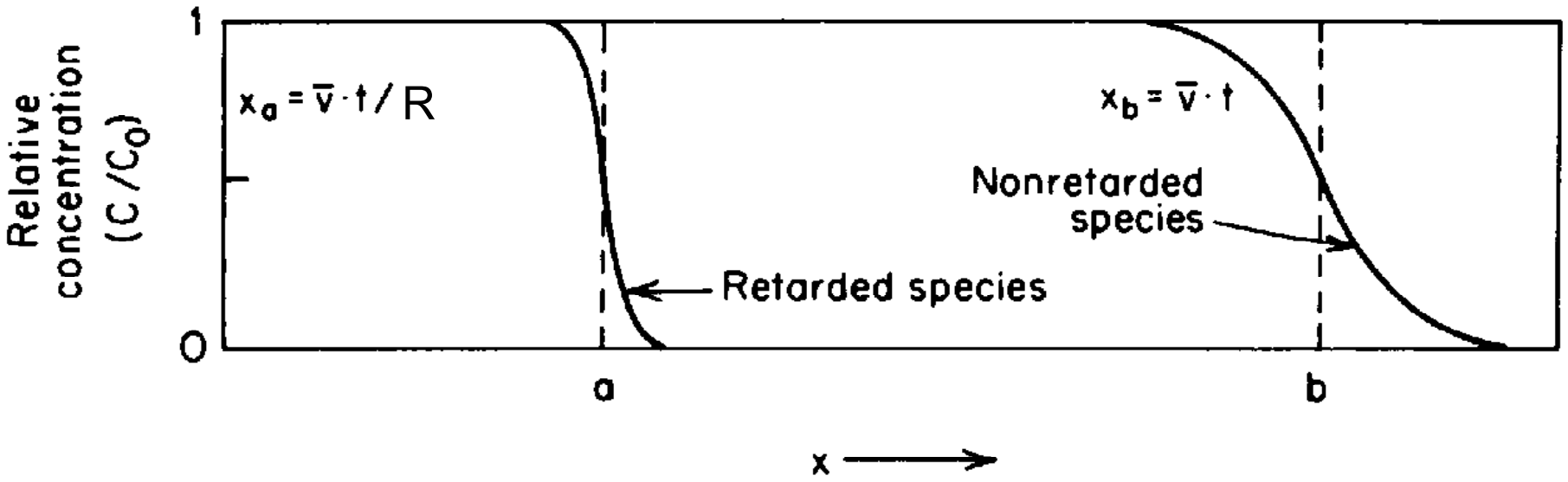
Retardation in the field



Retardation =

delay in arrival of a compound relative to a conservative compound
(or relative to water flow velocity)

Retardation



$$R = \frac{v_g}{v_r}$$

R = Retardation factor (-)

v_g = Average linear velocity of groundwater flow (m s^{-1})

v_r = Velocity of the retarded compound (m s^{-1})

Retardation under Henry sorption

$$R = 1 + \frac{\rho_b}{n_e} K_d$$

$$K_d = \frac{c_s}{c_w}$$

R = Retardation factor (-)

K_d = Distribution coefficient soil-water (L kg^{-1})

ρ_b = Bulk mass density (kg/ L)

n_e = Effective porosity (-)

Retardation under Freundlich sorption

$$R = 1 + \frac{\rho_b}{n_e} n K_{fr} c^{n-1}$$

$$c_s = K_{fr} c_w^{\,n}$$

R = Retardation factor (-)

K_{fr} = Freundlich distribution coefficient soil-water $(L\ kg^{-1})^{1/n}$

n = Freundlich exponent (-)

c = Concentration of compound in water ($\mu g\ L^{-1}$)

ρ_b = Bulk mass density ($L\ kg^{-1}$)

n_e = Effective porosity (-)